

If $N_1/N_2 \ll 1$ and $N_3/N_4 \ll 1$, then the effect of J_2 and J_3 on the equivalent moment of inertia J_{1eq} is negligible. Similar comments apply to the equivalent viscous-friction coefficient b_{1eq} of the gear train. In terms of the equivalent moment of inertia J_{1eq} and equivalent viscous-friction coefficient b_{1eq} , Equation (4-70) can be simplified to give

$$J_{1eq}\ddot{\theta}_1 + b_{1eq}\dot{\theta}_1 + nT_L = T_m$$

where

$$n = \frac{N_1}{N_2} \frac{N_3}{N_4}$$

- A-4-5.** Show that the torque-to-inertia ratios referred to the motor shaft and to the load shaft differ from each other by a factor of n . Show also that the torque squared-to-inertia ratios referred to the motor shaft and to the load shaft are the same.

Solution. Suppose that T_{max} is the maximum torque that can be produced on the motor shaft. Then the torque-to-inertia ratio referred to the motor shaft is

$$\frac{T_{max}}{J_m + n^2 J_L}$$

where J_m = moment of inertia of the rotor

J_L = moment of inertia of the load

n = gear ratio

The torque-to-inertia ratio referred to the load shaft is

$$\frac{\frac{T_{max}}{n}}{J_L + \frac{J_m}{n^2}} = \frac{nT_{max}}{J_m + n^2 J_L}$$

Clearly, they differ from each other by a factor of n . Hence, in comparing torque-to-inertia ratios of motors, we find it necessary to specify which shaft is the reference.

Note that the ratio of torque squared to inertia referred to the motor shaft is

$$\frac{T_{max}^2}{J_m + n^2 J_L}$$

and that referred to the load shaft is

$$\frac{\frac{T_{max}^2}{n^2}}{J_L + \frac{J_m}{n^2}} = \frac{T_{max}^2}{J_m + n^2 J_L}$$

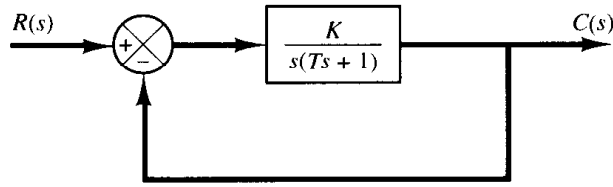
These two ratios are clearly the same.

- A-4-6.** When the system shown in Figure 4-41(a) is subjected to a unit-step input, the system output responds as shown in Figure 4-41(b). Determine the values of K and T from the response curve.

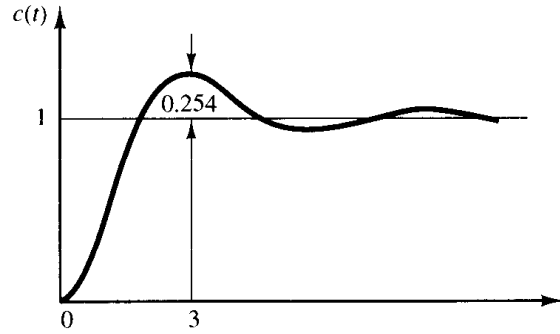
Solution. The maximum overshoot of 25.4% corresponds to $\zeta = 0.4$. From the response curve we have

$$t_p = 3$$

Consequently,



(a)



(b)

Figure 4-41

(a) Closed-loop system; (b) unit-step response curve.

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_n \sqrt{1 - 0.4^2}} = 3$$

It follows that

$$\omega_n = 1.14$$

From the block diagram we have

$$\frac{C(s)}{R(s)} = \frac{K}{Ts^2 + s + K}$$

from which

$$\omega_n = \sqrt{\frac{K}{T}}, \quad 2\zeta\omega_n = \frac{1}{T}$$

Therefore, the values of T and K are determined as

$$T = \frac{1}{2\zeta\omega_n} = \frac{1}{2 \times 0.4 \times 1.14} = 1.09$$

$$K = \omega_n^2 T = 1.14^2 \times 1.09 = 1.42$$

A-4-7. Determine the values of K and k of the closed-loop system shown in Figure 4-42 so that the maximum overshoot in unit-step response is 25% and the peak time is 2 sec. Assume that $J = 1 \text{ kg-m}^2$.

Solution. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{K}{Js^2 + Kks + K}$$

By substituting $J = 1 \text{ kg-m}^2$ into this last equation, we have

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + Kks + K}$$

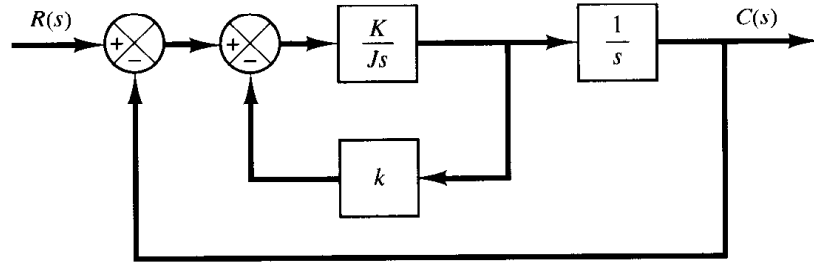


Figure 4-42
Closed-loop system.

Note that

$$\omega_n = \sqrt{K}, \quad 2\zeta\omega_n = Kk$$

The maximum overshoot M_p is

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

which is specified as 25%. Hence

$$e^{-\zeta\pi/\sqrt{1-\zeta^2}} = 0.25$$

from which

$$\frac{\zeta\pi}{\sqrt{1-\zeta^2}} = 1.386$$

or

$$\zeta = 0.404$$

The peak time t_p is specified as 2 sec. And so

$$t_p = \frac{\pi}{\omega_d} = 2$$

or

$$\omega_d = 1.57$$

Then the undamped natural frequency ω_n is

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = \frac{1.57}{\sqrt{1-0.404^2}} = 1.72$$

Therefore, we obtain

$$K = \omega_n^2 = 1.72^2 = 2.95 \text{ N-m}$$

$$k = \frac{2\zeta\omega_n}{K} = \frac{2 \times 0.404 \times 1.72}{2.95} = 0.471 \text{ sec}$$

A-4-8. What is the unit-step response of the system shown in Figure 4-43?

Solution. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{10s + 10}{s^2 + 10s + 10}$$

For the unit-step input $[R(s) = 1/s]$, we have

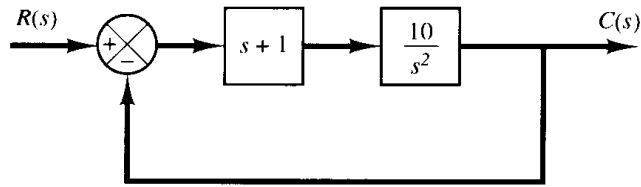


Figure 4-43
Closed-loop system.

$$\begin{aligned}
 C(s) &= \frac{10s + 10}{s^2 + 10s + 10} \frac{1}{s} \\
 &= \frac{10s + 10}{(s + 5 + \sqrt{15})(s + 5 - \sqrt{15})s} \\
 &= \frac{-4 - \sqrt{15}}{3 + \sqrt{15}} \frac{1}{s + 5 + \sqrt{15}} + \frac{-4 + \sqrt{15}}{3 - \sqrt{15}} \frac{1}{s + 5 - \sqrt{15}} + \frac{1}{s}
 \end{aligned}$$

The inverse Laplace transform of $C(s)$ gives

$$\begin{aligned}
 c(t) &= -\frac{4 + \sqrt{15}}{3 + \sqrt{15}} e^{-(5+\sqrt{15})t} + \frac{4 - \sqrt{15}}{-3 + \sqrt{15}} e^{-(5-\sqrt{15})t} + 1 \\
 &= -1.1455e^{-8.87t} + 0.1455e^{-1.13t} + 1
 \end{aligned}$$

Clearly, the output will not exhibit any oscillation. The response curve exponentially approaches the final value $c(\infty) = 1$.

- A-4-9.** Figure 4-44(a) shows a mechanical vibratory system. When 2 lb of force (step input) is applied to the system, the mass oscillates, as shown in Figure 4-44(b). Determine m , b , and k of the system from this response curve. The displacement x is measured from the equilibrium position.

Solution. The transfer function of this system is

$$\frac{X(s)}{P(s)} = \frac{1}{ms^2 + bs + k}$$

Since

$$P(s) = \frac{2}{s}$$

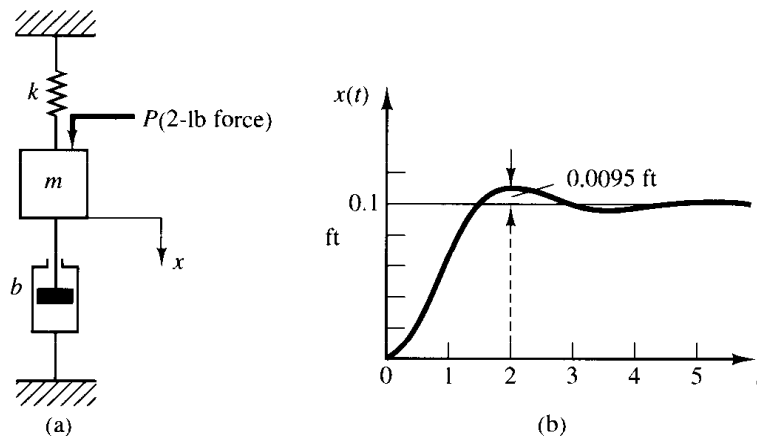


Figure 4-44
(a) Mechanical vibratory system; (b) step-response curve.

we obtain

$$X(s) = \frac{2}{s(ms^2 + bs + k)}$$

It follows that the steady-state value of x is

$$x(\infty) = \lim_{s \rightarrow 0} sX(s) = \frac{2}{k} = 0.1 \text{ ft}$$

Hence

$$k = 20 \text{ lb}_f/\text{ft}$$

Note that $M_p = 9.5\%$ corresponds to $\zeta = 0.6$. The peak time t_p is given by

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{0.8\omega_n}$$

The experimental curve shows that $t_p = 2 \text{ sec}$. Therefore,

$$\omega_n = \frac{3.14}{2 \times 0.8} = 1.96 \text{ rad/sec}$$

Since $\omega_n^2 = k/m = 20/m$, we obtain

$$m = \frac{20}{\omega_n^2} = \frac{20}{1.96^2} = 5.2 \text{ slugs} = 166 \text{ lb}$$

(Note that 1 slug = 1 lb_f-sec²/ft.) Then b is determined from

$$2\zeta\omega_n = \frac{b}{m}$$

or

$$b = 2\zeta\omega_n m = 2 \times 0.6 \times 1.96 \times 5.2 = 12.2 \text{ lb}_f/\text{ft/sec}$$

A-4-10. Assuming that the mechanical system shown in Figure 4-45 is at rest before excitation force $P \sin \omega t$ is given, derive the complete solution $x(t)$ and the steady-state solution $x_{ss}(t)$. The displacement x is measured from the equilibrium position. Assume that the system is underdamped.

Solution. The equation of motion for the system is

$$m\ddot{x} + b\dot{x} + kx = P \sin \omega t$$

Noting that $x(0) = 0$ and $\dot{x}(0) = 0$, the Laplace transform of this equation is

$$(ms^2 + bs + k)X(s) = P \frac{\omega}{s^2 + \omega^2}$$

or

$$X(s) = \frac{P\omega}{(s^2 + \omega^2)(ms^2 + bs + k)}$$

Since the system is underdamped, $X(s)$ can be written as follows:

$$X(s) = \frac{P\omega}{m} \frac{1}{s^2 + \omega^2} \frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad \text{where } 0 < \zeta < 1$$

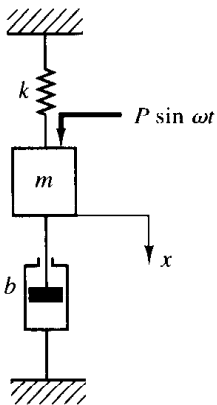


Figure 4-45
Mechanical system.

where $\omega_n = \sqrt{k/m}$ and $\zeta = b/(2\sqrt{mk})$. $X(s)$ can be expanded as

$$X(s) = \frac{P\omega}{m} \left(\frac{as + c}{s^2 + \omega^2} + \frac{-as + d}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right)$$

By simple calculations it can be found that

$$a = \frac{-2\zeta\omega_n}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}, \quad c = \frac{(\omega_n^2 - \omega^2)}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}, \quad d = \frac{4\zeta^2\omega_n^2 - (\omega_n^2 - \omega^2)}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}$$

Hence

$$X(s) = \frac{P\omega}{m} \frac{1}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2} \left[\frac{-2\zeta\omega_n s + (\omega_n^2 - \omega^2)}{s^2 + \omega^2} + \frac{2\zeta\omega_n(s + \zeta\omega_n) + 2\zeta^2\omega_n^2 - (\omega_n^2 - \omega^2)}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right]$$

The inverse Laplace transform of $X(s)$ gives

$$x(t) = \frac{P\omega}{m[(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2]} \left[-2\zeta\omega_n \cos \omega t + \frac{(\omega_n^2 - \omega^2)}{\omega} \sin \omega t + 2\zeta\omega_n e^{-\zeta\omega_n t} \cos \omega_n \sqrt{1 - \zeta^2} t + \frac{2\zeta^2\omega_n^2 - (\omega_n^2 - \omega^2)}{\omega_n \sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1 - \zeta^2} t \right]$$

At steady state ($t \rightarrow \infty$) the terms involving $e^{-\zeta\omega_n t}$ approach zero. Thus at steady state

$$\begin{aligned} x_{ss}(t) &= \frac{P\omega}{m[(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2]} \left(-2\zeta\omega_n \cos \omega t + \frac{\omega_n^2 - \omega^2}{\omega} \sin \omega t \right) \\ &= \frac{P\omega}{(k - m\omega^2)^2 + b^2\omega^2} \left(-b \cos \omega t + \frac{k - m\omega^2}{\omega} \sin \omega t \right) \\ &= \frac{P}{\sqrt{(k - m\omega^2)^2 + b^2\omega^2}} \sin \left(\omega t - \tan^{-1} \frac{b\omega}{k - m\omega^2} \right) \end{aligned}$$

A-4-11. Consider the unit-step response of the second-order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The amplitude of the exponentially damped sinusoid changes as a geometric series. At time $t = t_p = \pi/\omega_d$, the amplitude is equal to $e^{-(\sigma/\omega_d)\pi}$. After one oscillation, or at $t = t_p + 2\pi/\omega_d = 3\pi/\omega_d$, the amplitude is equal to $e^{-(\sigma/\omega_d)3\pi}$; after another cycle of oscillation, the amplitude is $e^{-(\sigma/\omega_d)5\pi}$. The logarithm of the ratio of successive amplitudes is called the *logarithmic decrement*. Determine the logarithmic decrement for this second-order system. Describe a method for experimental determination of the damping ratio from the rate of decay of the oscillation.

Solution. Let us define the amplitude of the output oscillation at $t = t_i$ to be x_i , where $t_i = t_p + (i - 1)T$ (T = period of oscillation). The amplitude ratio per one period of damped oscillation is

$$\frac{x_1}{x_2} = \frac{e^{-(\sigma/\omega_d)\pi}}{e^{-(\sigma/\omega_d)3\pi}} = e^{2(\sigma/\omega_d)\pi} = e^{2\zeta\pi/\sqrt{1-\zeta^2}}$$

Thus, the logarithmic decrement δ is

$$\delta = \ln \frac{x_1}{x_2} = \frac{2\zeta\pi}{\sqrt{1-\zeta^2}}$$

It is a function only of the damping ratio ζ . Thus, the damping ratio ζ can be determined by use of the logarithmic decrement.

In the experimental determination of the damping ratio ζ from the rate of decay of the oscillation, we measure the amplitude x_1 at $t = t_p$ and amplitude x_n at $t = t_p + (n-1)T$. Note that it is necessary to choose n large enough so that the ratio x_1/x_n is not near unity. Then

$$\frac{x_1}{x_n} = e^{(n-1)2\zeta\pi/\sqrt{1-\zeta^2}}$$

or

$$\ln \frac{x_1}{x_n} = (n-1) \frac{2\zeta\pi}{\sqrt{1-\zeta^2}}$$

Hence

$$\zeta = \frac{\frac{1}{n-1} \left(\ln \frac{x_1}{x_n} \right)}{\sqrt{4\pi^2 + \left[\frac{1}{n-1} \left(\ln \frac{x_1}{x_n} \right) \right]^2}}$$

- A-4-12.** In the system shown in Figure 4-46, the numerical values of m , b , and k are given as $m = 1$ kg, $b = 2$ N-sec/m, and $k = 100$ N/m. The mass is displaced 0.05 m and released without initial velocity. Find the frequency observed in the vibration. In addition, find the amplitude four cycles later. The displacement x is measured from the equilibrium position.

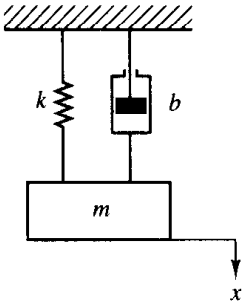


Figure 4-46
Spring-mass-damper system.

Solution. The equation of motion for the system is

$$m\ddot{x} + b\dot{x} + kx = 0$$

Substituting the numerical values for m , b , and k into this equation gives

$$\ddot{x} + 2\dot{x} + 100x = 0$$

where the initial conditions are $x(0) = 0.05$ and $\dot{x}(0) = 0$. From this last equation the undamped natural frequency ω_n and the damping ratio ζ are found to be

$$\omega_n = 10, \quad \zeta = 0.1$$

The frequency actually observed in the vibration is the damped natural frequency ω_d .

$$\omega_d = \omega_n \sqrt{1-\zeta^2} = 10\sqrt{1-0.01} = 9.95 \text{ rad/sec}$$

In the present analysis, $\dot{x}(0)$ is given as zero. Thus, solution $x(t)$ can be written as

$$x(t) = x(0)e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right)$$

It follows that at $t = nT$, where $T = 2\pi/\omega_d$,

$$x(nT) = x(0)e^{-\zeta\omega_n nT}$$

Consequently, the amplitude four cycles later becomes

$$\begin{aligned} x(4T) &= x(0)e^{-\zeta\omega_n 4T} = x(0)e^{-(0.1)(10)(4)(0.6315)} \\ &= 0.05e^{-2.526} = 0.05 \times 0.07998 = 0.004 \text{ m} \end{aligned}$$

- A-4-13.** Consider a system whose closed-loop poles and closed-loop zero are located in the s plane on a line parallel to the $j\omega$ axis, as shown in Figure 4-47. Show that the impulse response of such a system is a damped cosine function.

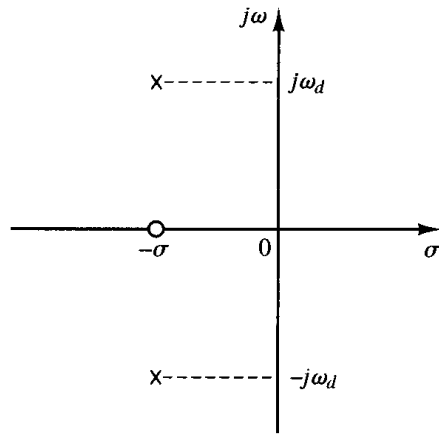


Figure 4-47
Closed-loop pole-zero configuration of system whose impulse response is a damped cosine function.

Solution. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{K(s + \sigma)}{(s + \sigma + j\omega_d)(s + \sigma - j\omega_d)}$$

For a unit-impulse input, $R(s) = 1$ and

$$C(s) = \frac{K(s + \sigma)}{(s + \sigma)^2 + \omega_d^2}$$

The inverse Laplace transform of $C(s)$ is

$$c(t) = Ke^{-\sigma t} \cos \omega_d t, \quad \text{for } t \geq 0$$

which is a damped cosine function.

- A-4-14.** Consider the liquid-level control system shown in Figure 4-48. The controller is of the proportional type. The set point of the controller is fixed.

Draw a block diagram of the system, assuming that changes in the variables are small. Obtain the transfer function between the level of the second tank and the disturbance input q_d . Obtain the steady-state error when the disturbance q_d is a unit-step function.

Solution. Figure 4-49(a) is a block diagram of this system when changes in the variables are small. Since the set point of the controller is fixed, $r = 0$. (Note that r is the change in set point.)

To investigate the response of the level of the second tank subjected to a unit-step disturbance q_d , we find it convenient to modify the block diagram of Figure 4-49(a) to the one shown in Figure 4-49(b).

The transfer function between $H_2(s)$ and $Q_d(s)$ can be obtained as

Figure 4-48
Liquid-level control
system.

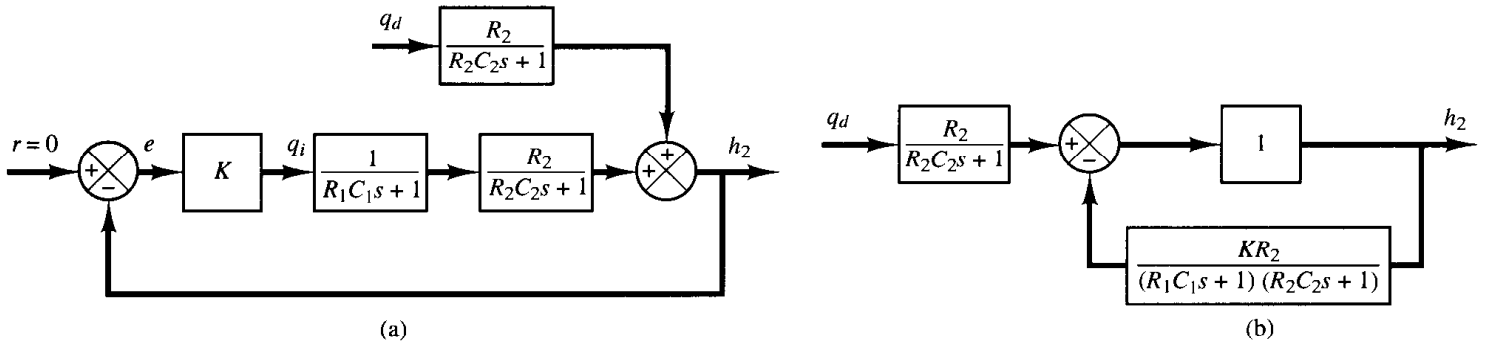
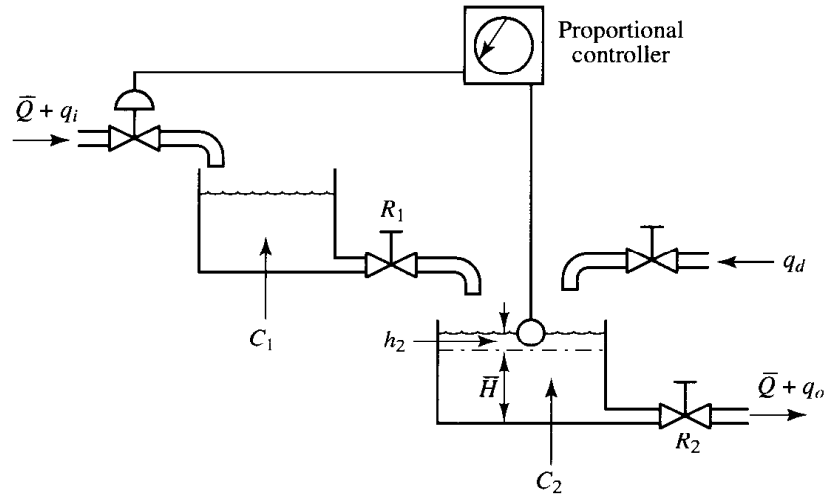


Figure 4-49

(a) Block diagram of the system shown in Figure 4-48; (b) modified block diagram.

$$\frac{H_2(s)}{Q_d(s)} = \frac{R_2(R_1 C_1 s + 1)}{(R_1 C_1 s + 1)(R_2 C_2 s + 1) + K R_2}$$

From this equation, the response $H_2(s)$ to the disturbance $Q_d(s)$ can be found. The effect of the controller is seen by the presence of K in the denominator of this last equation.

For the unit-step disturbance $Q_d(s)$, we obtain

$$h_2(\infty) = \frac{R_2}{1 + K R_2}$$

or

$$\text{Steady-state error} = -\frac{R_2}{1 + K R_2}$$

The system exhibits offset in the response to a unit-step disturbance.

Note that the characteristic equation for the disturbance input and that for the reference input are the same. The characteristic equation for this system is

$$(R_1 C_1 s + 1)(R_2 C_2 s + 1) + K R_2 = 0$$

which can be modified to

$$s^2 + \left(\frac{R_1 C_1 + R_2 C_2}{R_1 C_1 R_2 C_2} \right) s + \frac{1 + K R_2}{R_1 C_1 R_2 C_2} = 0$$

The undamped natural frequency ω_n and the damping ratio ζ are given by

$$\omega_n = \sqrt{\frac{1 + K R_2}{R_1 C_1 R_2 C_2}}, \quad \zeta = \frac{R_1 C_1 + R_2 C_2}{2 \sqrt{R_1 C_1 R_2 C_2} \sqrt{1 + K R_2}}$$

Both the undamped natural frequency and the damping ratio depend on the value of the gain K . This gain must be adjusted so that the transient responses to both the reference input and disturbance input show reasonable damping and reasonable speed.

- A-4-15.** Consider the liquid-level control system shown in Figure 4-50. The inlet valve is controlled by a hydraulic integral controller. Assume that the steady-state inflow rate is $\bar{Q}$ and steady-state outflow rate is also $\bar{Q}$, the steady-state head is $\bar{H}$, steady-state pilot valve displacement is $\bar{X} = 0$, and steady-state valve position is $\bar{Y}$. We assume that the set point $\bar{R}$ corresponds to the steady-state head $\bar{H}$. The set point is fixed. Assume also that the disturbance inflow rate q_d , which is a small quantity, is applied to the water tank at $t = 0$. This disturbance causes the head to change from $\bar{H}$ to $\bar{H} + h$. This change results in a change in the outflow rate by q_o . Through the hydraulic controller, the change in head causes a change in the inflow rate from $\bar{Q}$ to $\bar{Q} + q_i$. (The integral controller tends to keep the head constant as much as possible in the presence of disturbances.) We assume that all changes are of small quantities.

Assuming the following numerical values for the system,

$$\begin{aligned} C &= 2 \text{ m}^2, & R &= 0.5 \text{ sec/m}^2, & K_v &= 1 \text{ m}^2/\text{sec} \\ a &= 0.25 \text{ m}, & b &= 0.75 \text{ m}, & K_1 &= 4 \text{ sec}^{-1} \end{aligned}$$

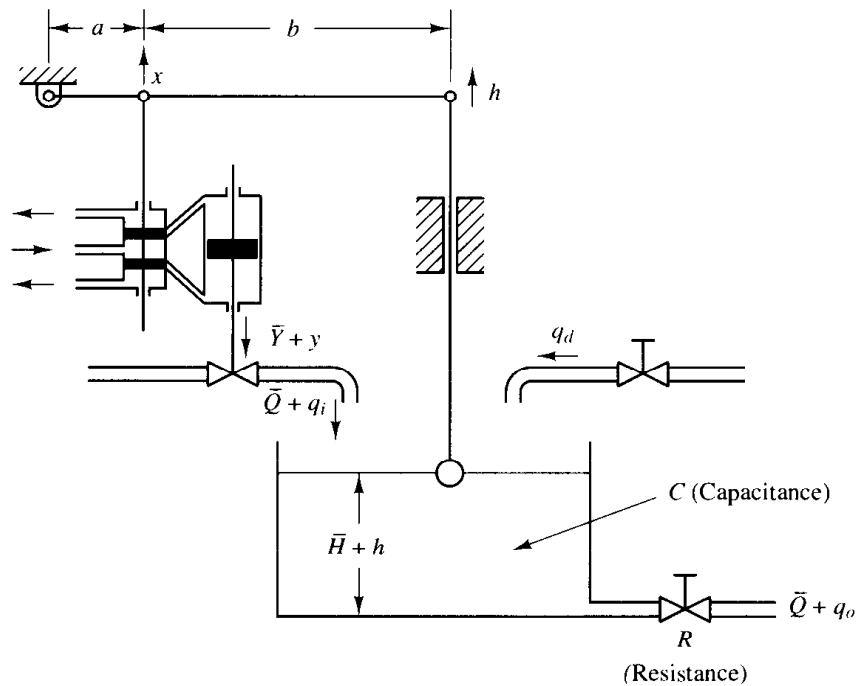


Figure 4-50
Liquid-level control system.

obtain the response $h(t)$ when the disturbance input q_d is a unit-step function. Also obtain this response $h(t)$ with MATLAB.

Solution. Since the increase of water in the tank during dt seconds is equal to the net inflow to the tank during the same dt seconds, we have

$$C dh = (q_i - q_o + q_d) dt \quad (4-71)$$

where

$$q_o = \frac{h}{R} \quad (4-72)$$

For the feedback lever mechanism, we have

$$x = \frac{a}{a+b} h \quad (4-73)$$

We assume that the velocity of the power piston (valve) is proportional to pilot valve displacement x , or

$$\frac{dy}{dt} = K_1 x \quad (4-74)$$

where K_1 is a positive constant. We also assume that the change in the inflow rate q_i is negatively proportional to the change in the valve opening y , or

$$q_i = -K_v y \quad (4-75)$$

where K_v is a positive constant.

Now we obtain the equations for the system as follows: From Equations (4-71), (4-72), and (4-75), we get

$$C \frac{dh}{dt} = -K_v y - \frac{h}{R} + q_d \quad (4-76)$$

From Equations (4-73) and (4-74), we have

$$\frac{dy}{dt} = \frac{K_1 a}{a+b} h \quad (4-77)$$

By substituting the given numerical values into Equations (4-76) and (4-77), we obtain

$$2 \frac{dh}{dt} = -y - 2h + q_d$$

$$\frac{dy}{dt} = h$$

Taking the Laplace transforms of the preceding two equations, assuming zero initial conditions, we obtain

$$2sH(s) = -Y(s) - 2H(s) + Q_d(s)$$

$$sY(s) = H(s)$$

By eliminating $Y(s)$ from the last two equations and noting that the disturbance input is a unit-step function, or $Q_d(s) = 1/s$, we get

$$H(s) = \frac{s}{2s^2 + 2s + 1} \frac{1}{s} = \frac{0.5}{(s + 0.5)^2 + 0.5^2}$$

The inverse Laplace transform of $H(s)$ gives the time response $h(t)$.

$$h(t) = e^{-0.5t} \sin 0.5t$$

Notice that the unit-step disturbance input q_d caused a transient error in the head which becomes zero at steady state. The integral controller thus eliminated the error caused by the disturbance input q_d .

Plotting the response curve $h(t)$ with MATLAB. Since the response $H(s)$ is given by

$$H(s) = \frac{s}{2s^2 + 2s + 1} \frac{1}{s}$$

MATLAB Program 4-13 may be used to obtain the response to the unit-step disturbance input. The resulting response curve is shown in Figure 4-51.

MATLAB Program 4-13

```
% ----- Unit-step response -----

% ***** Enter numerator and denominator of the transfer
% function *****

num = [0 1 0];
den = [2 2 1];

% ***** Enter step command *****

step(num,den)
grid
title('Unit-Step Response')
```

A-4-16. Consider the impulse response of the standard second-order system defined by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

For a unit-impulse input, $R(s) = 1$. Thus

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2 s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \frac{1}{s}$$

Consider the normalized system where $\omega_n = 1$. Then

$$C(s) = \frac{s}{s^2 + 2\zeta s + 1} \frac{1}{s}$$

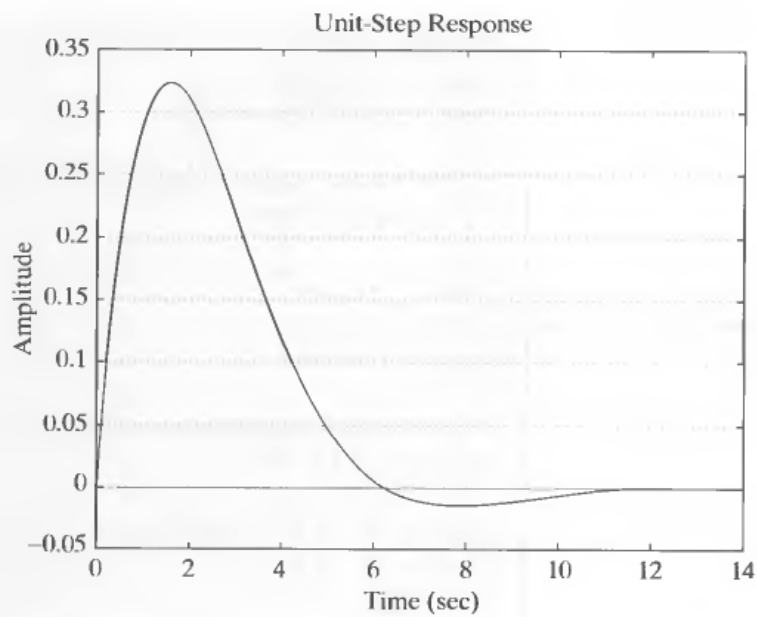


Figure 4-51
Response to unit-
step disturbance
input.

Consider five different values of zeta: $\zeta = 0.1, 0.3, 0.5, 0.7$, and 1.0 . Obtain the unit-impulse response curves for each zeta with MATLAB.

Solution. A MATLAB program for plotting the five unit-impulse response curves in one diagram is given in MATLAB Program 4-14. The resulting diagram is shown in Figure 4-52.

MATLAB Program 4-14

```
% ----- Unit-impulse response -----

% ***** Unit-impulse response curves for the normalized
% second-order system  $G(s) = 1/[s^2 + 2(\text{zeta})s + 1]$  *****

% ***** The unit-impulse response is obtained as the
% unit-step response of  $sG(s)$  *****

% ***** The values of zeta considered here are 0.1, 0.3,
% 0.5, 0.7, and 1.0 *****

% ***** Enter the numerator and denominator of  $sG(s)$  for
% zeta = 0.1 *****

num = [0 1 0];
den1 = [1 0.2 1];

% ***** Specify the computing time points (such as  $t = 0:0.1:10$ ).
% Then enter the step-response command step(num,den,t) and text
% command text(,') *****
```

```

t = 0:0.1:10;
step(num,den1,t);
text(2.2,0.88,'Zeta = 0.1')

% ***** Hold this plot and add other unit-impulse response
% curves to it *****

hold

Current plot held

% ***** Enter denominators of sG(s) for zeta = 0.3, 0.5,
% 0.7, and 1.0 *****

den2 = [1 0.6 1]; den3 = [1 1 1]; den4 = [1 1.4 1];
den5 = [1 2 1];

% ***** Superimpose on the held plot the unit-impulse response
% curves for zeta = 0.3, 0.5, 0.7, and 1.0 by entering
% successively the step-response command step(num,den,t)
% and text command text(,') *****

step(num,den2,t);
text(1.33,0.72,'0.3')
step(num,den3,t);
text(1.15,0.58,'0.5')
step(num,den4,t);
text(1.1,0.46,'0.7')
step(num,den5,t);
text(0.8,0.28,'1.0')

% ***** Enter grid and title of the plot *****

grid
title(Impulse-Response Curves for  $G(s) = 1/[s^2 + 2(\text{zeta})s + 1]$ )

% ***** Clear hold on graphics *****

hold

Current plot released

```

From the unit-impulse response curves for different values of zeta , we may conclude that if the impulse response $c(t)$ does not change sign the system is either critically damped or overdamped, in which case the corresponding step response does not overshoot, but increases or decreases monotonically and approaches a constant value.

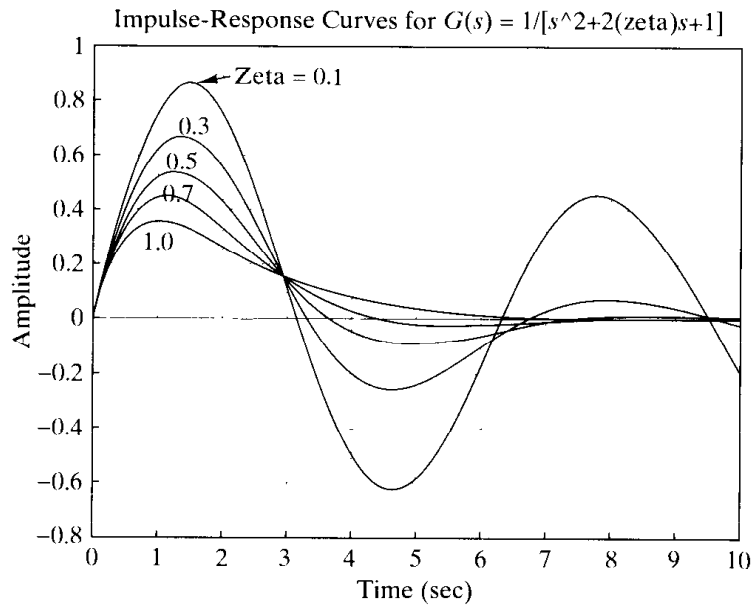


Figure 4-52
Unit-impulse re-
sponse curves.

PROBLEMS

B-4-1. A thermometer requires 1 min to indicate 98% of the response to a step input. Assuming the thermometer to be a first-order system, find the time constant.

If the thermometer is placed in a bath, the temperature of which is changing linearly at a rate of $10^\circ/\text{min}$, how much error does the thermometer show?

B-4-2. Consider the system shown in Figure 4-53. An armature-controlled dc servomotor drives a load consisting of the moment of inertia J_L . The torque developed by the motor is T . The angular displacements of the motor rotor and the load element are θ_m and θ , respectively. The gear ratio is $n = \theta/\theta_m$. Obtain the transfer function $\Theta(s)/E_i(s)$.

B-4-3. Consider the system shown in Figure 4-54(a). The damping ratio of this system is 0.158 and the undamped nat-

ural frequency is 3.16 rad/sec. To improve the relative stability, we employ tachometer feedback. Figure 4-54(b) shows such a tachometer-feedback system.

Determine the value of K_h so that the damping ratio of the system is 0.5. Draw unit-step response curves of both the original and tachometer-feedback systems. Also draw the error-versus-time curves for the unit-ramp response of both systems.

B-4-4. Obtain the unit-step response of a unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{4}{s(s + 5)}$$

B-4-5. Consider the unit-step response of a unity-feedback control system whose open-loop transfer function is

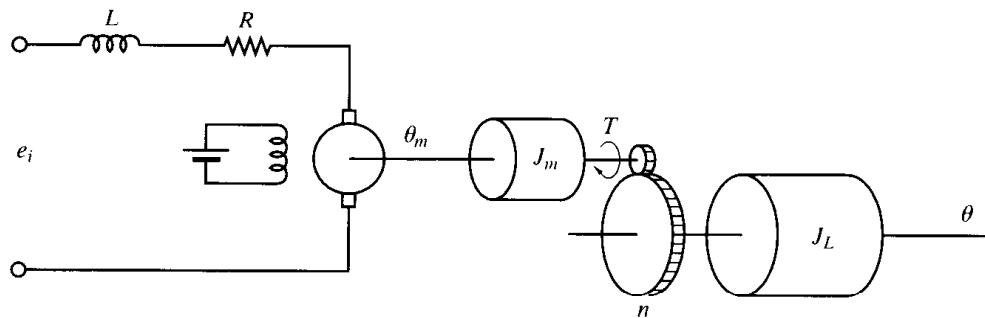
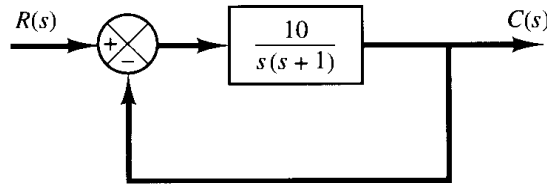
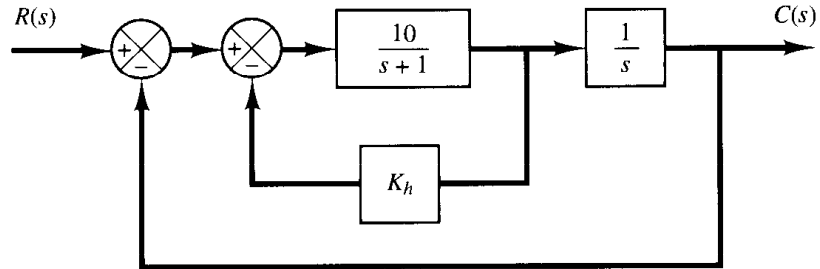


Figure 4-53
Armature-controlled
dc servomotor
system.



(a)



(b)

Figure 4-54

(a) Control system;
(b) control system
with tachometer
feedback.

$$G(s) = \frac{1}{s(s+1)}$$

Obtain the rise time, peak time, maximum overshoot, and settling time.

B-4-6. Consider the closed-loop system given by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Determine the values of ζ and ω_n so that the system responds to a step input with approximately 5% overshoot and with a settling time of 2 sec. (Use the 2% criterion.)

B-4-7. Figure 4-55 is a block diagram of a space-vehicle attitude-control system. Assuming the time constant T of the

controller to be 3 sec and the ratio of torque to inertia K/J to be $\frac{2}{9}$ rad²/sec², find the damping ratio of the system.

B-4-8. Consider the system shown in Figure 4-56. The system is initially at rest. Suppose that the cart is set into motion by an impulsive force whose strength is unity. Can it be stopped by another such impulsive force?

B-4-9. Obtain the unit-impulse response and the unit-step response of a unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{2s+1}{s^2}$$

B-4-10. Consider the system shown in Figure 4-57. Show that the transfer function $Y(s)/X(s)$ has a zero in the right-

Figure 4-55

Space-vehicle atti-
tude-control system.

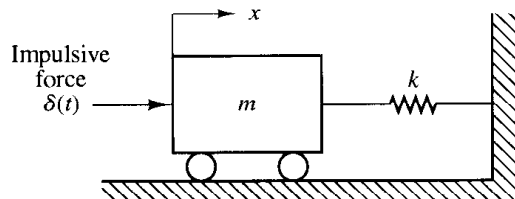
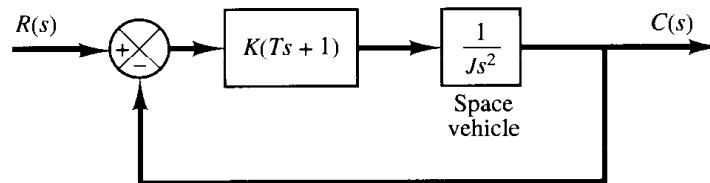


Figure 4-56

Mechanical system.

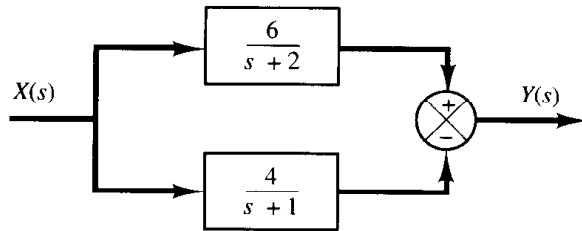


Figure 4-57
System with zero in the right-half s plane.

half s plane. Then obtain $y(t)$ when $x(t)$ is a unit step. Plot $y(t)$ versus t .

B-4-11. An oscillatory system is known to have a transfer function of the following form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Assume that a record of a damped oscillation is available as shown in Figure 4-58. Determine the damping ratio ζ of the system from the graph.

B-4-12. Referring to the system shown in Figure 4-59, determine the values of K and k such that the system has a damping ratio ζ of 0.7 and an undamped natural frequency ω_n of 4 rad/sec.

B-4-13. Consider the system shown in Figure 4-60. Determine the value of k such that the damping ratio ζ is 0.5. Then obtain the rise time t_r , peak time t_p , maximum overshoot M_p , and settling time t_s in the unit-step response.

B-4-14. Using MATLAB, obtain the unit-step response, unit-ramp response, and unit-impulse response of the following system:

$$\frac{C(s)}{R(s)} = \frac{10}{s^2 + 2s + 10}$$

where $R(s)$ and $C(s)$ are Laplace transforms of the input $r(t)$ and output $c(t)$, respectively.

B-4-15. Using MATLAB, obtain the unit-step response, unit-ramp response, and unit-impulse response of the

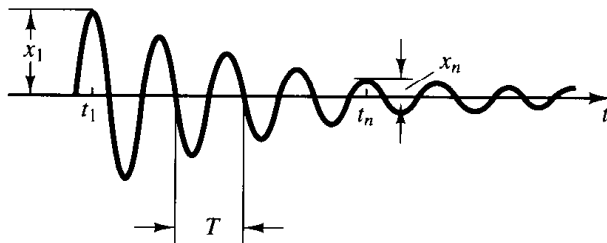


Figure 4-58
Decaying oscillation.

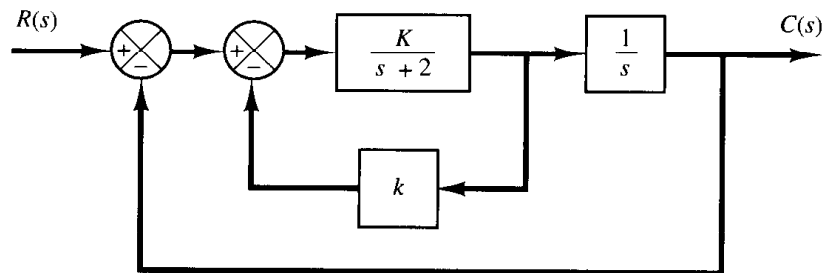


Figure 4-59
Closed-loop system.

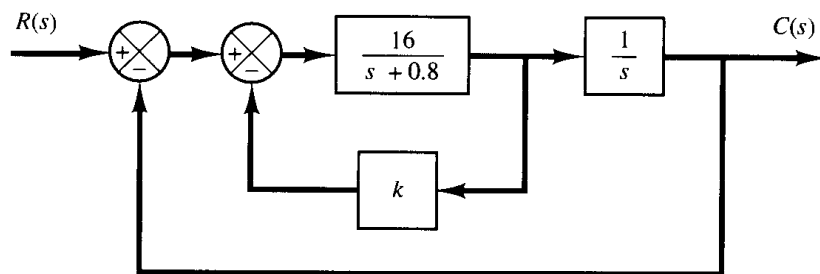


Figure 4-60
Block diagram of a system.

following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} u$$
$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where u is the input and y is the output.

B-4-16. Consider the same problem as discussed in Problem A-4-16. It is desired to use different marks for different curves (such as 'o', 'x', '--', '-', ':'). Modify MATLAB Program 4-14 for this purpose.

5

Basic Control Actions and Response of Control Systems

5-1 INTRODUCTION

An automatic controller compares the actual value of the plant output with the reference input (desired value), determines the deviation, and produces a control signal that will reduce the deviation to zero or to a small value. The manner in which the automatic controller produces the control signal is called the *control action*.

In this chapter we shall first discuss the basic control actions used in industrial control systems. Then we shall discuss the effects of integral and derivative control actions on the system response. We shall next consider the response of higher-order systems. Any physical system will become unstable if any of the closed-loop poles lies in the right-half s plane. To check the existence or nonexistence of such right-half plane poles, the Routh stability criterion is useful. We shall include discussions of this stability criterion in this chapter.

Many industrial automatic controllers are electronic, hydraulic, pneumatic, or their combinations. In this chapter we present principles of pneumatic controllers, hydraulic controllers, and electronic controllers.

The outline of this chapter follows: Section 5-1 has presented introductory material. Section 5-2 gives the basic control actions commonly used in industrial automatic controllers. Section 5-3 discusses the effects of integral and derivative control actions on system performance. Section 5-4 deals with higher-order systems, and Section 5-5 treats Routh's stability criterion. Sections 5-6 and 5-7 discuss pneumatic controllers and hydraulic controllers, respectively. Here we introduce the principle of operation of pneumatic and hydraulic controllers and methods for generating various control actions.

Section 5–8 treats electronic controllers using operational amplifiers. Section 5–9 discusses phase lead and phase lag in sinusoidal response. We derive the sinusoidal transfer function and show phase lead and phase lag that may occur in the sinusoidal response. Finally, in Section 5–10 we treat steady-state errors in system responses.

5-2 BASIC CONTROL ACTIONS

In this section we shall discuss the details of basic control actions used in industrial analog controllers. We shall begin with classifications of industrial analog controllers.

Classifications of industrial controllers. Industrial controllers may be classified according to their control actions as:

1. Two-position or on–off controllers
2. Proportional controllers
3. Integral controllers
4. Proportional-plus-integral controllers
5. Proportional-plus-derivative controllers
6. Proportional-plus-integral-plus-derivative controllers

Most industrial controllers use electricity or pressurized fluid such as oil or air as power sources. Controllers may also be classified according to the kind of power employed in the operation, such as pneumatic controllers, hydraulic controllers, or electronic controllers. What kind of controller to use must be decided based on the nature of the plant and the operating conditions, including such considerations as safety, cost, availability, reliability, accuracy, weight, and size.

Automatic controller, actuator, and sensor (measuring element). Figure 5–1 is a block diagram of an industrial control system, which consists of an automatic controller, an actuator, a plant, and a sensor (measuring element). The controller detects the actuating error signal, which is usually at a very low power level, and amplifies it to a sufficiently high level. The output of an automatic controller is fed to an actuator, such as a pneumatic motor or valve, a hydraulic motor, or an electric motor. (The actuator is

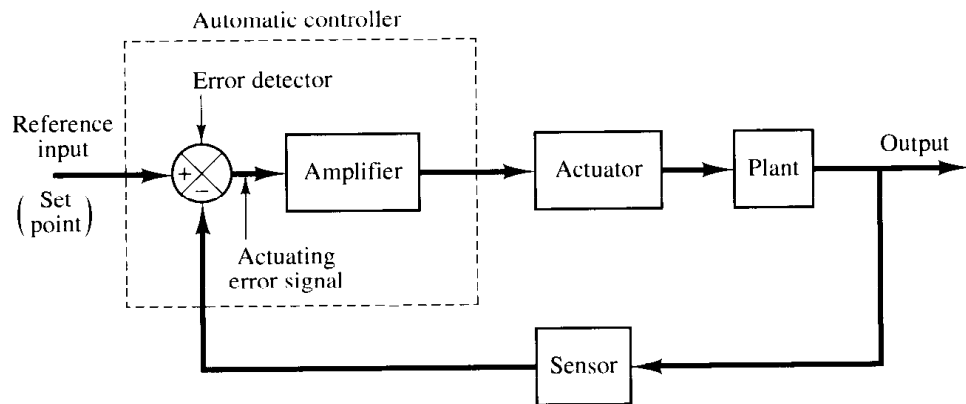


Figure 5–1
Block diagram of an industrial control system, which consists of an automatic controller, an actuator, a plant, and a sensor (measuring element).

a power device that produces the input to the plant according to the control signal so that the output signal will approach the reference input signal.)

The sensor or measuring element is a device that converts the output variable into another suitable variable, such as a displacement, pressure, or voltage, that can be used to compare the output to the reference input signal. This element is in the feedback path of the closed-loop system. The set point of the controller must be converted to a reference input with the same units as the feedback signal from the sensor or measuring element.

Self-operated controllers. In most industrial automatic controllers, separate units are used for the measuring element and for the actuator. In a very simple one, however, such as a self-operated controller, these elements are assembled in one unit. Self-operated controllers utilize power developed by the measuring element and are very simple and inexpensive. An example of such a self-operated controller is shown in Figure 5-2. The set point is determined by the adjustment of the spring force. The controlled pressure is measured by the diaphragm. The actuating error signal is the net force acting on the diaphragm. Its position determines the valve opening.

The operation of the self-operated controller is as follows: Suppose that the output pressure is lower than the reference pressure, as determined by the set point. Then the downward spring force is greater than the upward pressure force, resulting in a downward movement of the diaphragm. This increases the flow rate and raises the output pressure. When the upward pressure force equals the downward spring force, the valve plug stays stationary and the flow rate is constant. Conversely, if the output pressure is higher than the reference pressure, the valve opening becomes small and reduces the flow rate through the valve opening. Such a self-operated controller is widely used for water and gas pressure control.

Two-position or on-off control action. In a two-position control system, the actuating element has only two fixed positions, which are, in many cases, simply on and off. Two-position or on-off control is relatively simple and inexpensive and, for this reason, is very widely used in both industrial and domestic control systems.

Let the output signal from the controller be $u(t)$ and the actuating error signal be $e(t)$. In two-position control, the signal $u(t)$ remains at either a maximum or minimum value, depending on whether the actuating error signal is positive or negative, so that

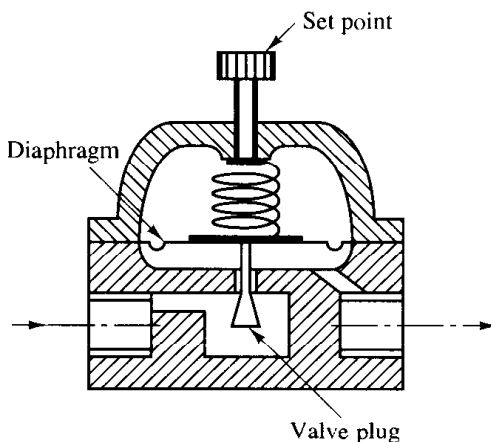


Figure 5-2
Self-operated controller.

$$u(t) = U_1, \quad \text{for } e(t) > 0$$

$$= U_2, \quad \text{for } e(t) < 0$$

where U_1 and U_2 are constants. The minimum value U_2 is usually either zero or $-U_1$. Two-position controllers are generally electrical devices, and an electric solenoid-operated valve is widely used in such controllers. Pneumatic proportional controllers with very high gains act as two-position controllers and are sometimes called pneumatic two-position controllers.

Figures 5-3 (a) and (b) show the block diagrams for two-position controllers. The range through which the actuating error signal must move before the switching occurs is called the *differential gap*. A differential gap is indicated in Figure 5-3(b). Such a differential gap causes the controller output $u(t)$ to maintain its present value until the actuating error signal has moved slightly beyond the zero value. In some cases, the differential gap is a result of unintentional friction and lost motion; however, quite often it is intentionally provided in order to prevent too frequent operation of the one-off mechanism.

Consider the liquid-level control system shown in Figure 5-4(a), where the electromagnetic valve shown in Figure 5-4(b) is used for controlling the inflow rate. This valve is either open or closed. With this two-position control, the water inflow rate is either a positive constant or zero. As shown in Figure 5-5, the output signal continuously moves between the two limits required to cause the actuating element to move from one fixed

Figure 5-3

(a) Block diagram of an on-off controller;
(b) block diagram of an on-off controller with differential gap.

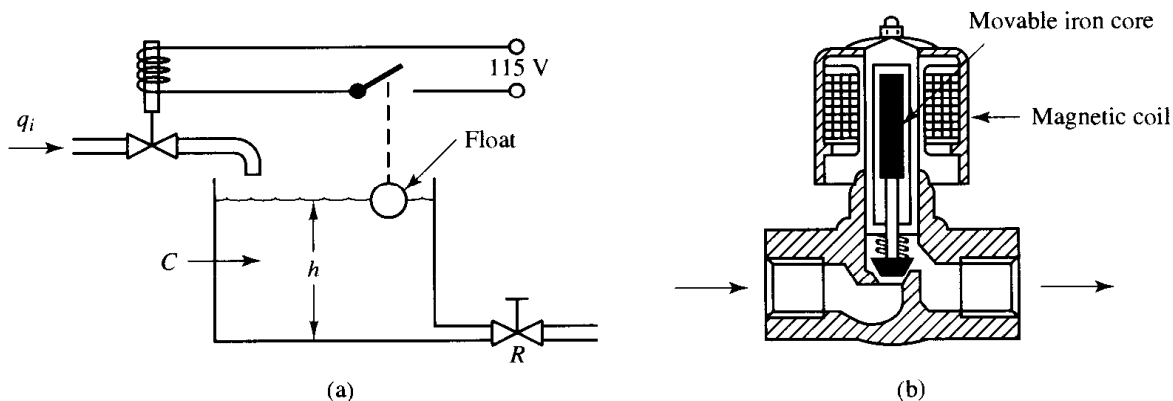
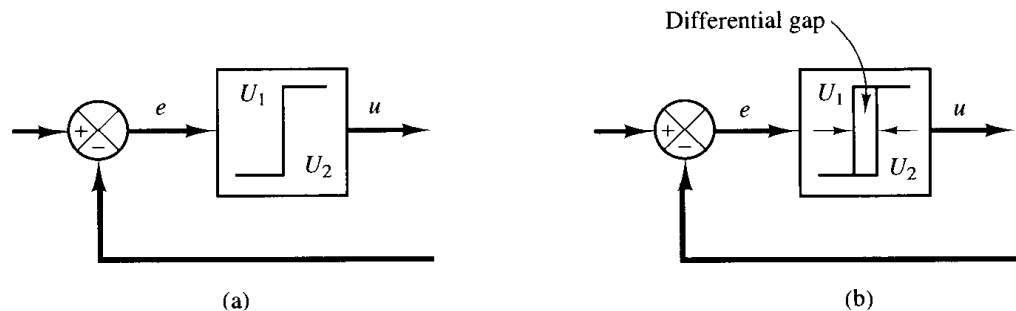


Figure 5-4

(a) Liquid-level control system; (b) electromagnetic valve.

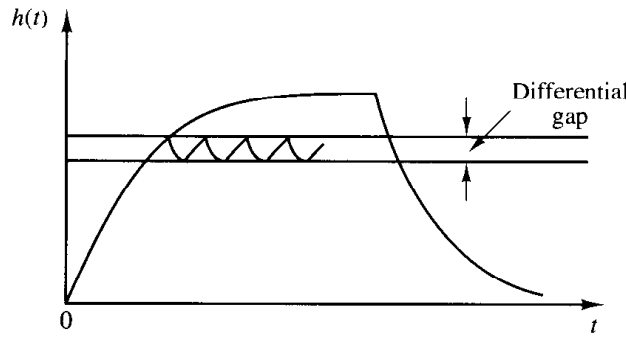


Figure 5-5
Level $h(t)$ versus t curve for the system shown in Figure 5-4(a).

position to the other. Notice that the output curve follows one of two exponential curves, one corresponding to the filling curve and the other to the emptying curve. Such output oscillation between two limits is a typical response characteristic of a system under two-position control.

From Figure 5-5, we notice that the amplitude of the output oscillation can be reduced by decreasing the differential gap. The decrease in the differential gap, however, increases the number of on-off switchings per minute and reduces the useful life of the component. The magnitude of the differential gap must be determined from such considerations as the accuracy required and the life of the component.

Proportional control action. For a controller with proportional control action, the relationship between the output of the controller $u(t)$ and the actuating error signal $e(t)$ is

$$u(t) = K_p e(t)$$

or, in Laplace-transformed quantities,

$$\frac{U(s)}{E(s)} = K_p$$

where K_p is termed the proportional gain.

Whatever the actual mechanism may be and whatever the form of the operating power, the proportional controller is essentially an amplifier with an adjustable gain. A block diagram of such a controller is shown in Figure 5-6.

Integral control action. In a controller with integral control action, the value of the controller output $u(t)$ is changed at a rate proportional to the actuating error signal $e(t)$. That is,

$$\frac{du(t)}{dt} = K_i e(t)$$

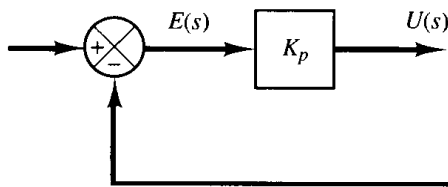


Figure 5-6
Block diagram of a proportional controller.

or

$$u(t) = K_i \int_0^t e(t) dt$$

where K_i is an adjustable constant. The transfer function of the integral controller is

$$\frac{U(s)}{E(s)} = \frac{K_i}{s}$$

If the value of $e(t)$ is doubled, then the value of $u(t)$ varies twice as fast. For zero actuating error, the value of $u(t)$ remains stationary. The integral control action is sometimes called reset control. Figure 5–7 shows a block diagram of such a controller.

Proportional-plus-integral control action. The control action of a proportional-plus-integral controller is defined by

$$u(t) = K_p e(t) + \frac{K_p}{T_i} \int_0^t e(t) dt$$

or the transfer function of the controller is

$$\frac{U(s)}{E(s)} = K_p \left(1 + \frac{1}{T_i s} \right)$$

where K_p is the proportional gain, and T_i is called the *integral time*. Both K_p and T_i are adjustable. The integral time adjusts the integral control action, while a change in the value of K_p affects both the proportional and integral parts of the control action. The inverse of the integral time T_i is called the *reset rate*. The reset rate is the number of times per minute that the proportional part of the control action is duplicated. Reset rate is measured in terms of repeats per minute. Figure 5–8 (a) shows a block diagram of a proportional-plus-integral controller. If the actuating error signal $e(t)$ is a unit-step function as shown in Figure 5–8(b), then the controller output $u(t)$ becomes as shown in Figure 5–8(c).

Proportional-plus-derivative control action. The control action of a proportional-plus-derivative controller is defined by

$$u(t) = K_p e(t) + K_p T_d \frac{de(t)}{dt}$$

and the transfer function is

$$\frac{U(s)}{E(s)} = K_p (1 + T_d s)$$

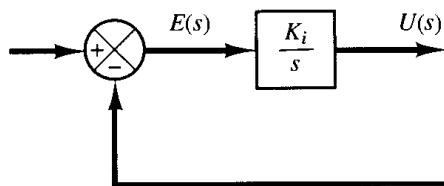


Figure 5–7
Block diagram of an integral controller.

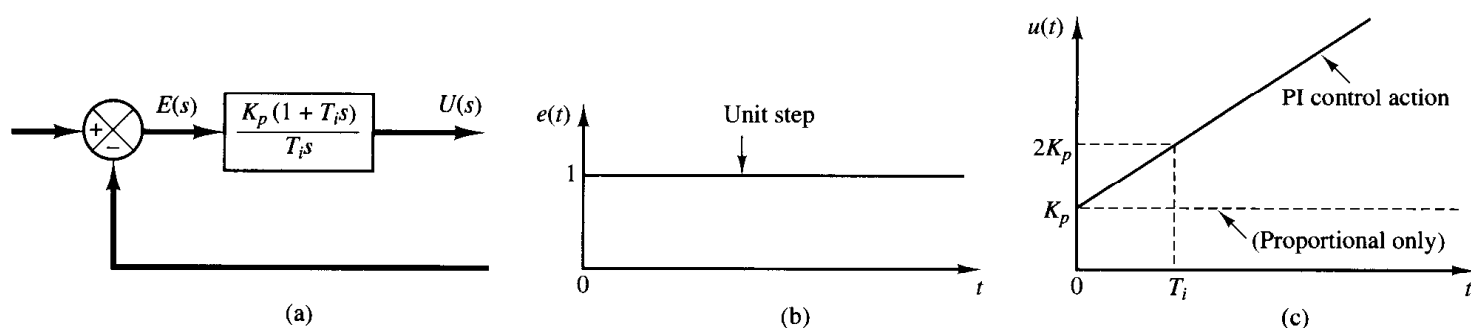


Figure 5-8

(a) Block diagram of a proportional-plus-integral controller; (b) and (c) diagrams depicting a unit-step input and the controller output.

where K_p is the proportional gain and T_d is a constant called the *derivative time*. Both K_p and T_d are adjustable. The derivative control action, sometimes called *rate control*, is where the magnitude of the controller output is proportional to the rate of change of the actuating error signal. The derivative time T_d is the time interval by which the rate action advances the effect of the proportional control action. Figure 5-9(a) shows a block diagram of a proportional-plus-derivative controller. If the actuating error signal $e(t)$ is a unit-ramp function as shown in Figure 5-9(b), then the controller output $u(t)$ becomes as shown in Figure 5-9(c). As may be seen from Figure 5-9(c), the derivative control action has an anticipatory character. As a matter of course, however, derivative control action can never anticipate any action that has not yet taken place.

While derivative control action has the advantage of being anticipatory, it has the disadvantages that it amplifies noise signals and may cause a saturation effect in the actuator.

Note that derivative control action can never be used alone because this control action is effective only during transient periods.

Proportional-plus-integral-plus-derivative control action. The combination of proportional control action, integral control action, and derivative control action is termed proportional-plus-integral-plus-derivative control action. This combined action

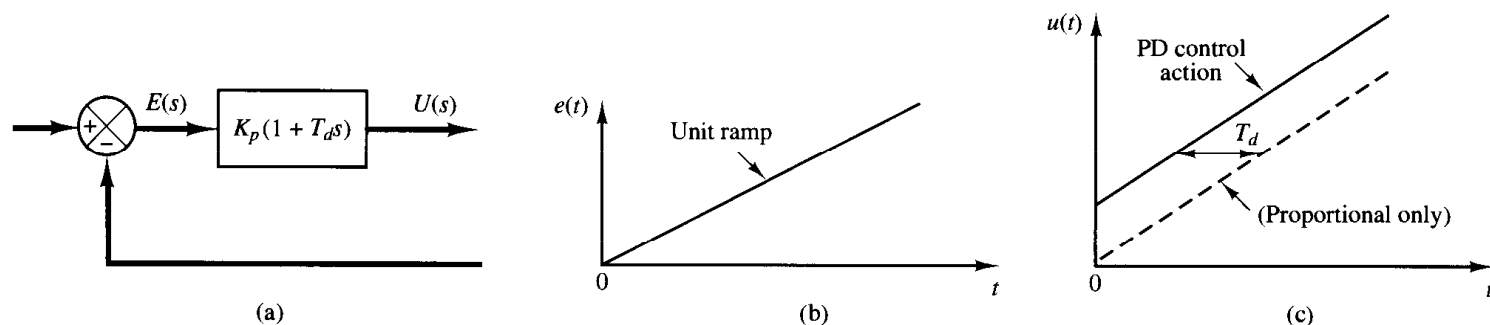


Figure 5-9

(a) Block diagram of a proportional-plus-derivative controller; (b) and (c) diagrams depicting a unit-ramp input and the controller output.

has the advantages of each of the three individual control actions. The equation of a controller with this combined action is given by

$$u(t) = K_p e(t) + \frac{K_p}{T_i} \int_0^t e(t) dt + K_p T_d \frac{de(t)}{dt}$$

or the transfer function is

$$\frac{U(s)}{E(s)} = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

where K_p is the proportional gain, T_i is the integral time, and T_d is the derivative time. The block diagram of a proportional-plus-integral-plus-derivative controller is shown in Figure 5–10(a). If $e(t)$ is a unit-ramp function as shown in Figure 5–10(b), then the controller output $u(t)$ becomes as shown in Figure 5–10(c).

Effects of the sensor (measuring element) on system performance. Since the dynamic and static characteristics of the sensor or measuring element affect the indication of the actual value of the output variable, the sensor plays an important role in determining the overall performance of the control system. The sensor usually determines the transfer function in the feedback path. If the time constants of a sensor are negligibly small compared with other time constants of the control system, the transfer function of the sensor simply becomes a constant. Figures 5–11(a), (b), and (c) show block diagrams of automatic controllers having a first-order sensor, an overdamped second-order sensor, and an underdamped second-order sensor, respectively. The response of a thermal sensor is often of the overdamped second-order type.

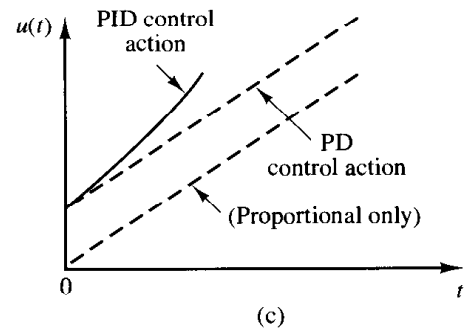
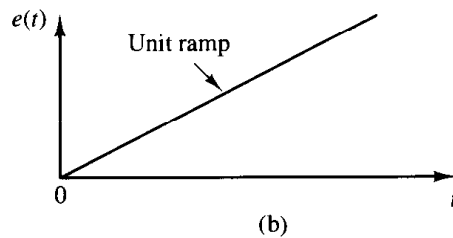
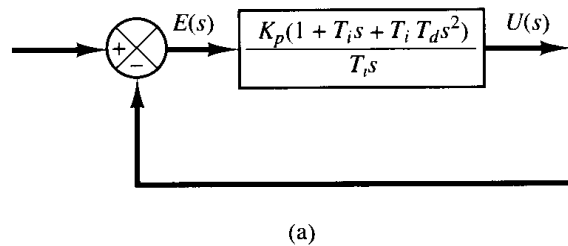


Figure 5–10

(a) Block diagram of a proportional-plus-integral-plus-derivative controller; (b) and (c) diagrams depicting a unit-ramp input and the controller output.

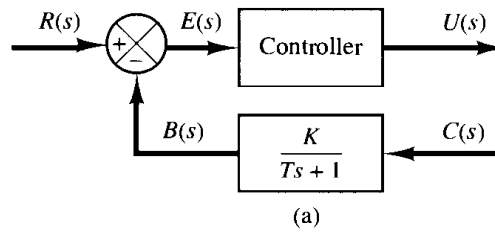
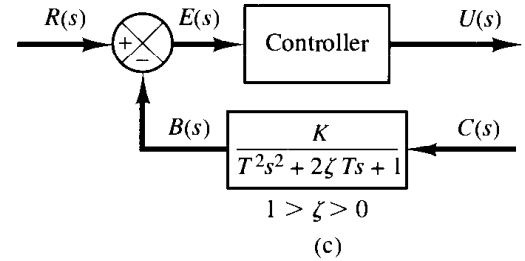
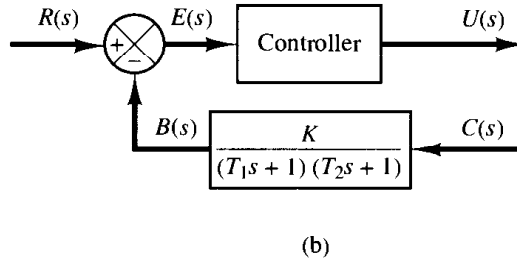


Figure 5–11

Block diagrams of automatic controllers with (a) first-order sensor; (b) over-damped second-order sensor; (c) underdamped second-order sensor.



5-3 EFFECTS OF INTEGRAL AND DERIVATIVE CONTROL ACTIONS ON SYSTEM PERFORMANCE

In this section, we shall investigate the effects of integral and derivative control actions on the system performance. Here we shall consider only simple systems so that the effects of integral and derivative control actions on system performance can be clearly seen.

Integral control action. In the proportional control of a plant whose transfer function does not possess an integrator $1/s$, there is a steady-state error, or offset, in the response to a step input. Such an offset can be eliminated if the integral control action is included in the controller.

In the integral control of a plant, the control signal, the output signal from the controller, at any instant is the area under the actuating error signal curve up to that instant. The control signal $u(t)$ can have a nonzero value when the actuating error signal $e(t)$ is zero, as shown in Figure 5–12(a). This is impossible in the case of the proportional controller since a nonzero control signal requires a nonzero actuating error signal. (A nonzero actuating error signal at steady state means that there is an offset.) Figure 5–12(b) shows the curve $e(t)$ versus t and the corresponding curve $u(t)$ versus t when the controller is of the proportional type.

Note that integral control action, while removing offset or steady-state error, may lead to oscillatory response of slowly decreasing amplitude or even increasing amplitude, both of which are usually undesirable.

Integral control of liquid-level control systems. In Section 4–2, we found that the proportional control of a liquid-level system will result in a steady-state error with a step input. We shall now show that such an error can be eliminated if integral control action is included in the controller.

Figure 5–12

(a) Plots of $e(t)$ and $u(t)$ curves showing nonzero control signal when the actuating error signal is zero (integral control); (b) plots of $e(t)$ and $u(t)$ curves showing zero control signal when the actuating error signal is zero (proportional control).

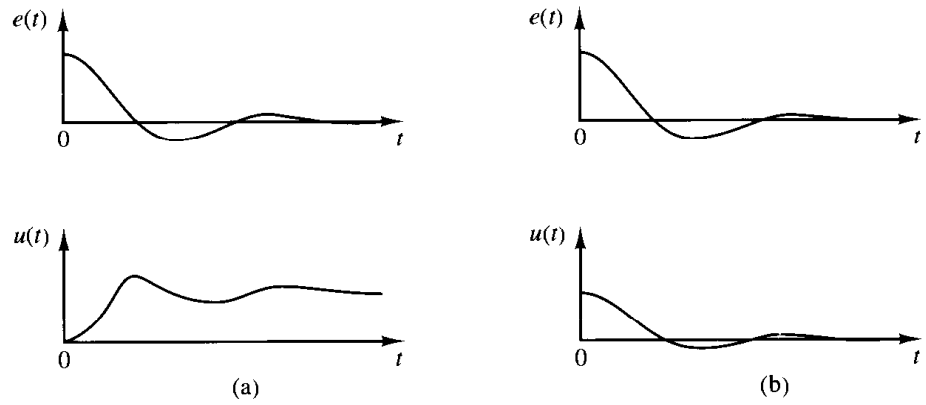


Figure 5–13(a) shows a liquid-level control system. We assume that the controller is an integral controller. We also assume that the variables x , q_i , h , and q_o , which are measured from their respective steady-state values $\bar{X}$, $\bar{Q}$, $\bar{H}$, and $\bar{Q}$ are small quantities so that the system can be considered linear. Under these assumptions, the block diagram of the system can be obtained as shown in Figure 5–13(b). From Figure 5–13(b), the closed-loop transfer function between $H(s)$ and $X(s)$ is

$$\frac{H(s)}{X(s)} = \frac{KR}{RCs^2 + s + KR}$$

Hence

$$\begin{aligned} \frac{E(s)}{X(s)} &= \frac{X(s) - H(s)}{X(s)} \\ &= \frac{RCs^2 + s}{RCs^2 + s + KR} \end{aligned}$$

Since the system is stable, the steady-state error for the unit-step response is obtained by applying the final-value theorem, as follows:

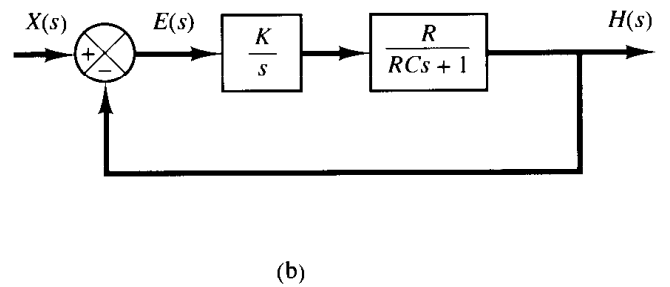
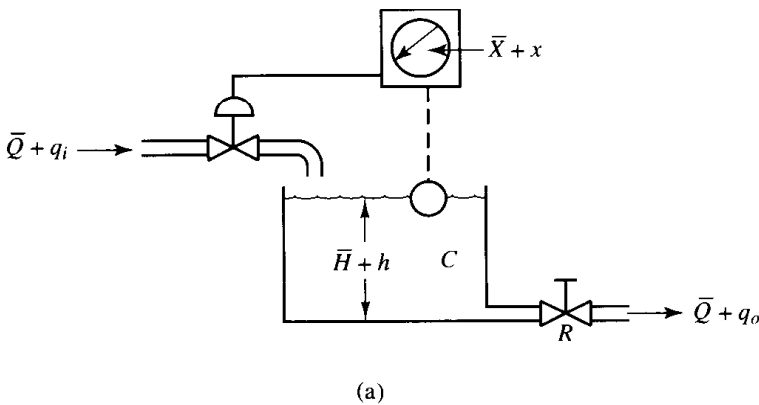


Figure 5–13

(a) Liquid-level control system; (b) block diagram of the system.

$$\begin{aligned}
e_{ss} &= \lim_{s \rightarrow 0} sE(s) \\
&= \lim_{s \rightarrow 0} \frac{s(RCs^2 + s)}{RCs^2 + s + KR} \frac{1}{s} \\
&= 0
\end{aligned}$$

Integral control of the liquid-level system thus eliminates the steady-state error in the response to the step input. This is an important improvement over the proportional control alone, which gives offset.

Response to torque disturbances (proportional control). Let us investigate the effect of a torque disturbance occurring at the load element. Consider the system shown in Figure 5–14. The proportional controller delivers torque T to position the load element, which consists of moment of inertia and viscous friction. Torque disturbance is denoted by D .

Assuming that the reference input is zero or $R(s) = 0$, the transfer function between $C(s)$ and $D(s)$ is given by

$$\frac{C(s)}{D(s)} = \frac{1}{Js^2 + bs + K_p}$$

Hence

$$\frac{E(s)}{D(s)} = -\frac{C(s)}{D(s)} = -\frac{1}{Js^2 + bs + K_p}$$

The steady-state error due to a step disturbance torque of magnitude T_d is given by

$$\begin{aligned}
e_{ss} &= \lim_{s \rightarrow 0} sE(s) \\
&= \lim_{s \rightarrow 0} \frac{-s}{Js^2 + bs + K_p} \frac{T_d}{s} \\
&= -\frac{T_d}{K_p}
\end{aligned}$$

At steady state, the proportional controller provides the torque $-T_d$, which is equal in magnitude but opposite in sign to the disturbance torque T_d . The steady-state output due to the step disturbance torque is

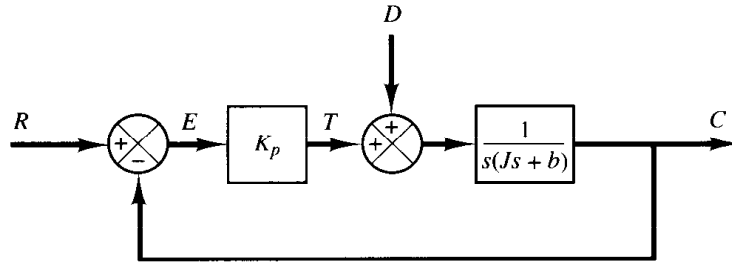


Figure 5–14
Control system with
a torque disturbance.

$$c_{ss} = -e_{ss} = \frac{T_d}{K_p}$$

The steady-state error can be reduced by increasing the value of the gain K_p . Increasing this value, however, will cause the system response to be more oscillatory.

Obtaining responses with MATLAB. In what follows, we shall obtain the response curves of the system shown in Figure 5–14 when it is subjected to a unit-step disturbance. Specifically, we shall obtain step-response curves for a small value of K_p and a large value of K_p .

Consider two cases:

Case 1: $J = 1, b = 0.5, K_p = 1$ (system 1):

$$\frac{C(s)}{D(s)} = \frac{1}{s^2 + 0.5s + 1}$$

Case 2: $J = 1, b = 0.5, K_p = 4$ (system 2):

$$\frac{C(s)}{D(s)} = \frac{1}{s^2 + 0.5s + 4}$$

Note that for system 1

$$\begin{aligned} \text{num1} &= [0 \quad 0 \quad 1] \\ \text{den1} &= [1 \quad 0.5 \quad 1] \end{aligned}$$

For system 2

$$\begin{aligned} \text{num2} &= [0 \quad 0 \quad 1] \\ \text{den2} &= [1 \quad 0.5 \quad 4] \end{aligned}$$

In MATLAB Program 5–1 we have used notations $y1$ and $y2$ for the response. $y1$ is the response $c(t)$ of system 1, and $y2$ is the response $c(t)$ of system 2.

In MATLAB Program 5–1, note that we have used the `plot` command with multiple arguments, rather than using the `hold` command. (We get the same result either way.) To use the `plot` command with multiple arguments, the sizes of the $y1$ and $y2$ vectors need not be the same. However, it is convenient if the two vectors are of the same length. Hence, we specify the same number of computing points by specifying the computing time points (such as $t = 0:0.1:20$). The `step` command must include this user-specified time t . Thus, in MATLAB Program 5–1 we have used the following `step` command:

$$[y, x, t] = \text{step}(\text{num}, \text{den}, t)$$

The unit-step response curves obtained by use of MATLAB Program 5–1 are shown in Figure 5–15.

MATLAB Program 5-1

```
% ----- Plotting two step-response curves on one
% diagram -----

% *****Enter numerators and denominators of two
% transfer functions*****

num1 = [0 0 1];
den1 = [1 0.5 1];
num2 = [0 0 1];
den2 = [1 0.5 4];

% ***** To plot two step-response curves y1 versus t
% and y2 versus t on one diagram and write texts
% 'System 1' and 'System 2' to distinguish two curves,
% enter the following commands *****

t = 0:0.1:20;
[y1,x1,t] = step(num1,den1,t);
[y2,x2,t] = step(num2,den2,t);
plot(t,y1,t,y2)
grid
text(11,0.75,'System 1'), text(11.2,0.16,'System 2')

% ***** Add title of the plot, xlabel, and ylabel *****

title('Step Responses of Two Systems')
xlabel('t Sec')
ylabel('Outputs y1 and y2')
```

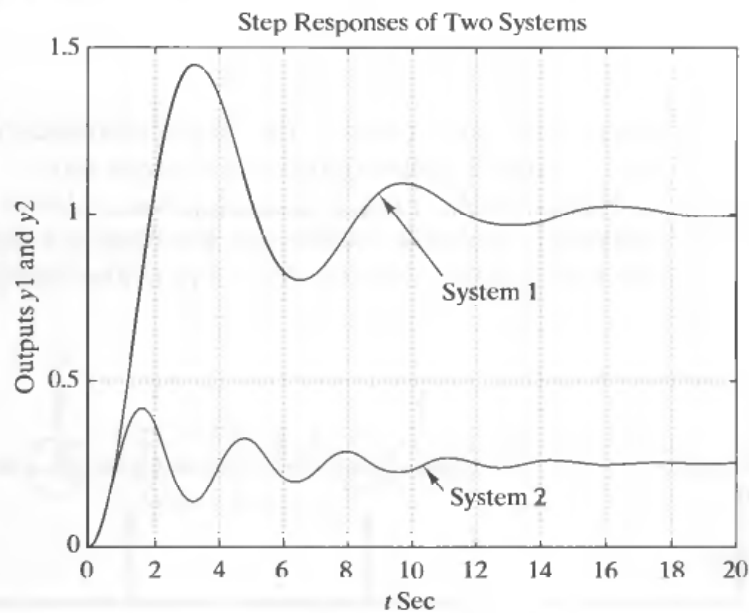


Figure 5-15
Unit-step response
curves.

Response to torque disturbances (proportional-plus-integral control). To eliminate offset due to torque disturbance, the proportional controller may be replaced by a proportional-plus-integral controller.

If integral control action is added to the controller, then, as long as there is an error signal, a torque is developed by the controller to reduce this error, provided the control system is a stable one.

Figure 5-16 shows the proportional-plus-integral control of the load element, consisting of moment of inertia and viscous friction.

The closed-loop transfer function between $C(s)$ and $D(s)$ is

$$\frac{C(s)}{D(s)} = \frac{s}{Js^3 + bs^2 + K_p s + \frac{K_p}{T_i}}$$

In the absence of the reference input, or $r(t) = 0$, the error signal is obtained from

$$E(s) = -\frac{s}{Js^3 + bs^2 + K_p s + \frac{K_p}{T_i}} D(s)$$

If this control system is stable, that is, if the roots of the characteristic equation

$$Js^3 + bs^2 + K_p s + \frac{K_p}{T_i} = 0$$

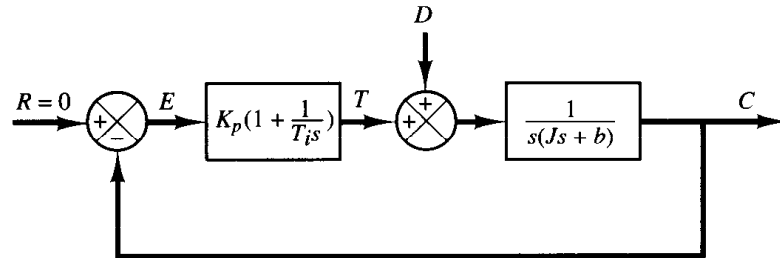
have negative real parts, then the steady-state error in the response to a unit-step disturbance torque can be obtained by applying the final-value theorem as follows:

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} sE(s) \\ &= \lim_{s \rightarrow 0} \frac{-s^2}{Js^3 + bs^2 + K_p s + \frac{K_p}{T_i}} \frac{1}{s} \\ &= 0 \end{aligned}$$

Thus steady-state error to the step disturbance torque can be eliminated if the controller is of the proportional-plus-integral type.

Note that the integral control action added to the proportional controller has converted the originally second-order system to a third-order one. Hence the control system may become unstable for a large value of K_p since the roots of the characteristic

Figure 5-16
Proportional-plus-integral control of a load element consisting of moment of inertia and viscous friction.



equation may have positive real parts. (The second-order system is always stable if the coefficients in the system differential equation are all positive.)

It is important to point out that if the controller were an integral controller, as in Figure 5–17, then the system always becomes unstable because the characteristic equation

$$Js^3 + bs^2 + K = 0$$

will have roots with positive real parts. Such an unstable system cannot be used in practice.

Note that in the system of Figure 5–16 the proportional control action tends to stabilize the system, while the integral control action tends to eliminate or reduce steady-state error in response to various inputs.

Derivative control action. Derivative control action, when added to a proportional controller, provides a means of obtaining a controller with high sensitivity. An advantage of using derivative control action is that it responds to the rate of change of the actuating error and can produce a significant correction before the magnitude of the actuating error becomes too large. Derivative control thus anticipates the actuating error, initiates an early corrective action, and tends to increase the stability of the system.

Although derivative control does not affect the steady-state error directly, it adds damping to the system and thus permits the use of a larger value of the gain K , which will result in an improvement in the steady-state accuracy.

Because derivative control operates on the rate of change of the actuating error and not the actuating error itself, this mode is never used alone. It is always used in combination with proportional or proportional-plus-integral control action.

Proportional control of systems with inertia load. Before we discuss the effect of derivative control action on system performance, we shall consider the proportional control of an inertia load.

Consider the system shown in Figure 5–18(a). The closed-loop transfer function is obtained as

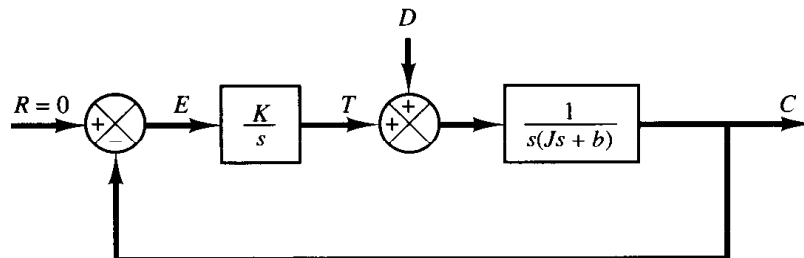
$$\frac{C(s)}{R(s)} = \frac{K_p}{Js^2 + K_p}$$

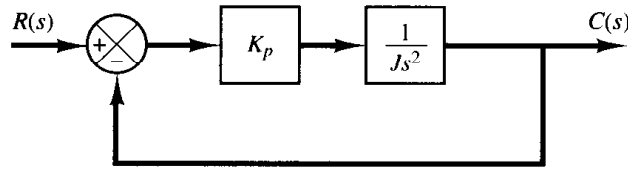
Since the roots of the characteristic equation

$$Js^2 + K_p = 0$$

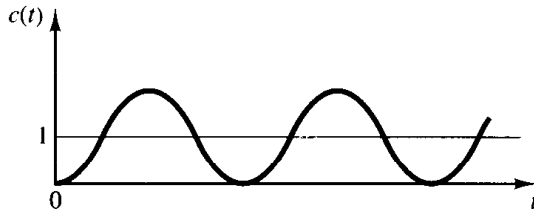
are imaginary, the response to a unit-step input continues to oscillate indefinitely, as shown in Figure 5–18(b).

Figure 5–17
Integral control of a load element consisting of moment of inertia and viscous friction.





(a)



(b)

Figure 5-18

(a) Proportional control of a system with inertia load; (b) response to a unit-step input.

Control systems exhibiting such response characteristics are not desirable. We shall see that the addition of derivative control will stabilize the system.

Proportional-plus-derivative control of a system with inertia load. Let us modify the proportional controller to a proportional-plus-derivative controller whose transfer function is $K_p(1 + T_d s)$. The torque developed by the controller is proportional to $K_p(e + T_d \dot{e})$. Derivative control is essentially anticipatory, measures the instantaneous error velocity, and predicts the large overshoot ahead of time and produces an appropriate counteraction before too large an overshoot occurs.

Consider the system shown in Figure 5-19(a). The closed-loop transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{K_p(1 + T_d s)}{Js^2 + K_p T_d s + K_p}$$

The characteristic equation

$$Js^2 + K_p T_d s + K_p = 0$$

now has two roots with negative real parts for positive values of J , K_p , and T_d . Thus derivative control introduces a damping effect. A typical response curve $c(t)$ to a unit-step input is shown in Figure 5-19(b). Clearly, the response curve shows a marked improvement over the original response curve shown in Figure 5-18(b).

Proportional-plus-derivative control of second-order systems. A compromise between acceptable transient-response behavior and acceptable steady-state behavior may be achieved by use of proportional-plus-derivative control action.

Consider the system shown in Figure 5-20. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{K_p + K_d s}{Js^2 + (B + K_d)s + K_p}$$

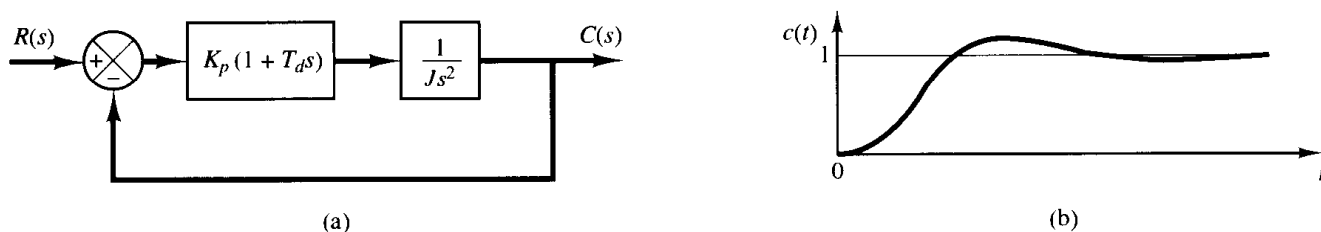


Figure 5-19

(a) Proportional-plus-derivative control of a system with inertia load; (b) response to a unit-step input.

The steady-state error for a unit-ramp input is

$$e_{ss} = \frac{B}{K_p}$$

The characteristic equation is

$$Js^2 + (B + K_d)s + K_p = 0$$

The effective damping coefficient of this system is thus $B + K_d$ rather than B . Since the damping ratio ζ of this system is

$$\zeta = \frac{B + K_d}{2\sqrt{K_p J}}$$

it is possible to make both the steady-state error e_{ss} for a ramp input and the maximum overshoot for a step input small by making B small, K_p large, and K_d large enough so that ζ is between 0.4 and 0.7.

In the following, we shall examine the unit-step response of the system shown in Figure 5-20. Let us define

$$\omega_n = \sqrt{\frac{K_p}{J}}, \quad z = \frac{K_p}{K_d}$$

The closed-loop transfer function can then be written

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{z} \frac{s + z}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

When a second-order system has a zero near the closed-loop poles, the transient-response behavior becomes considerably different from that of a second-order system without a zero.

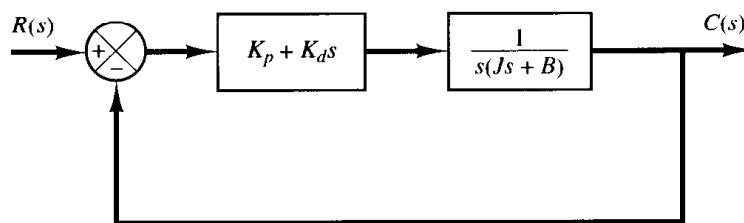


Figure 5-20
Control system.

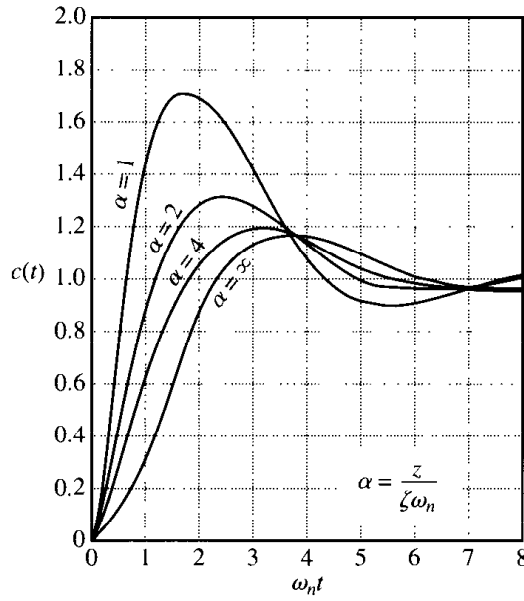


Figure 5-21
Unit-step response curves of the second-order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{z} \frac{s + z}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = 0.5$$

If the zero at $s = -z$ is located close to the $j\omega$ axis, the effect of the zero on the unit-step response is quite significant. Typical step-response curves of this system with $\zeta = 0.5$ and various values of $z/(\zeta\omega_n)$ are shown in Figure 5-21.

5-4 HIGHER-ORDER SYSTEMS

In this section, we shall first discuss the unit-step response of a particular type of higher-order system. We shall then present a transient response analysis of higher-order systems in general terms. Finally, we shall present a discussion of stability analysis in the complex plane.

Transient response of higher-order systems. Consider the system shown in Figure 5-22. The closed loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (5-1)$$

In general, $G(s)$ and $H(s)$ are given as ratios of polynomials in s , or

$$G(s) = \frac{p(s)}{q(s)} \quad \text{and} \quad H(s) = \frac{n(s)}{d(s)}$$

where $p(s)$, $q(s)$, $n(s)$, and $d(s)$ are polynomials in s . The closed-loop transfer function given by Equation (5-1) may then be written

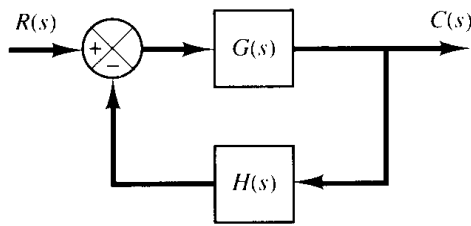


Figure 5-22
Control system.

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{p(s)d(s)}{q(s)d(s) + p(s)n(s)} \\ &= \frac{b_0s^m + b_1s^{m-1} + \cdots + b_{m-1}s + b_m}{a_0s^n + a_1s^{n-1} + \cdots + a_{n-1}s + a_n} \quad (m \leq n)\end{aligned}$$

The transient response of this system to any given input can be obtained by a computer simulation (see Section 4-4). If an analytical expression for the transient response is desired, then it is necessary to factor the denominator polynomial. [MATLAB may be used for finding the roots of the denominator polynomial. Use the command `roots(den)`.] Once the numerator and the denominator have been factored, $C(s)/R(s)$ can be written as

$$\frac{C(s)}{R(s)} = \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} \quad (5-2)$$

Let us examine the response behavior of this system to a unit-step input. Consider first the case where the closed-loop poles are all real and distinct. For a unit-step input, Equation (5-2) can be written

$$C(s) = \frac{a}{s} + \sum_{i=1}^n \frac{a_i}{s + p_i} \quad (5-3)$$

where a_i is the residue of the pole at $s = -p_i$.

If all closed-loop poles lie in the left-half s plane, the relative magnitudes of the residues determine the relative importance of the components in the expanded form of $C(s)$. If there is a closed-loop zero close to a closed-loop pole, then the residue at this pole is small and the coefficient of the transient-response term corresponding to this pole becomes small. A pair of closely located poles and zeros will effectively cancel each other. If a pole is located very far from the origin, the residue at this pole may be small. The transients corresponding to such a remote pole are small and last a short time. Terms in the expanded form of $C(s)$ having very small residues contribute little to the transient response, and these terms may be neglected. If this is done, the higher-order system may be approximated by a lower-order one. (Such an approximation often enables us to estimate the response characteristics of a higher-order system from those of a simplified one.)

Next, consider the case where the poles of $C(s)$ consist of real poles and pairs of complex-conjugate poles. A pair of complex-conjugate poles yields a second-order term in s . Since the factored form of the higher-order characteristic equation consists of first- and second-order terms, Equation (5-3) can be rewritten

$$C(s) = \frac{K \prod_{i=1}^m (s + z_i)}{s \prod_{j=1}^q (s + p_j) \prod_{k=1}^r (s^2 + 2\zeta_k \omega_k s + \omega_k^2)} \quad (5-4)$$

where $q + 2r = n$. If the closed-loop poles are distinct, Equation (5-4) can be expanded into partial fractions as follows:

$$C(s) = \frac{a}{s} + \sum_{j=1}^q \frac{a_j}{s + p_j} + \sum_{k=1}^r \frac{b_k(s + \zeta_k \omega_k) + c_k \omega_k \sqrt{1 - \zeta_k^2}}{s^2 + 2\zeta_k \omega_k s + \omega_k^2}$$

From this last equation, we see that the response of a higher-order system is composed of a number of terms involving the simple functions found in the responses of first- and second-order systems. The unit-step response $c(t)$, the inverse Laplace transform of $C(s)$, is then

$$c(t) = a + \sum_{j=1}^q a_j e^{-p_j t} + \sum_{k=1}^r b_k e^{-\zeta_k \omega_k t} \cos \omega_k \sqrt{1 - \zeta_k^2} t + \sum_{k=1}^r c_k e^{-\zeta_k \omega_k t} \sin \omega_k \sqrt{1 - \zeta_k^2} t, \quad \text{for } t \geq 0 \quad (5-5)$$

Thus the response curve of a stable higher-order system is the sum of a number of exponential curves and damped sinusoidal curves.

If all closed-loop poles lie in the left-half s plane, then the exponential terms and the damped exponential terms in Equation (5-5) will approach zero as time t increases. The steady-state output is then $c(\infty) = a$.

Let us assume that the system considered is a stable one. Then the closed-loop poles that are located far from the $j\omega$ axis have large negative real parts. The exponential terms that correspond to these poles decay very rapidly to zero. (Note that the horizontal distance from a closed-loop pole to the $j\omega$ axis determines the settling time of transients due to that pole. The smaller the distance is, the longer the settling time.)

Remember that the type of transient response is determined by the closed-loop poles, while the shape of the transient response is primarily determined by the closed-loop zeros. As we have seen earlier, the poles of the input $R(s)$ yield the steady-state response terms in the solution, while the poles of $C(s)/R(s)$ enter into the exponential transient-response terms and/or damped sinusoidal transient-response terms. The zeros of $C(s)/R(s)$ do not affect the exponents in the exponential terms, but they do affect the magnitudes and signs of the residues.

Dominant closed-loop poles. The relative dominance of closed-loop poles is determined by the ratio of the real parts of the closed-loop poles, as well as by the relative magnitudes of the residues evaluated at the closed-loop poles. The magnitudes of the residues depend on both the closed-loop poles and zeros.

If the ratios of the real parts exceed 5 and there are no zeros nearby, then the closed-loop poles nearest the $j\omega$ axis will dominate in the transient-response behavior because

these poles correspond to transient-response terms that decay slowly. Those closed-loop poles that have dominant effects on the transient-response behavior are called *dominant closed-loop* poles. Quite often the dominant closed-loop poles occur in the form of a complex-conjugate pair. The dominant closed-loop poles are most important among all closed-loop poles.

The gain of a higher-order system is often adjusted so that there will exist a pair of dominant complex-conjugate closed-loop poles. The presence of such poles in a stable system reduces the effect of such nonlinearities as dead zone, backlash, and coulomb friction.

Remember that, although the concept of dominant closed-loop poles is useful for estimating the dynamic behavior of a closed-loop system, we must be careful to see that the underlying assumptions are met before using it.

Stability analysis in the complex plane. The stability of a linear closed-loop system can be determined from the location of the closed-loop poles in the s plane. If any of these poles lie in the right-half s plane, then with increasing time they give rise to the dominant mode, and the transient response increases monotonically or oscillates with increasing amplitude. This represents an unstable system. For such a system, as soon as the power is turned on, the output may increase with time. If no saturation takes place in the system and no mechanical stop is provided, then the system may eventually be subjected to damage and fail since the response of a real physical system cannot increase indefinitely. Therefore, closed-loop poles in the right-half s plane are not permissible in the usual linear control system. If all closed-loop poles lie to the left of the $j\omega$ axis, any transient response eventually reaches equilibrium. This represents a stable system.

Whether a linear system is stable or unstable is a property of the system itself and does not depend on the input or driving function of the system. The poles of the input, or driving function, do not affect the property of stability of the system, but they contribute only to steady-state response terms in the solution. Thus, the problem of absolute stability can be solved readily by choosing no closed-loop poles in the right-half s plane, including the $j\omega$ axis. (Mathematically, closed-loop poles on the $j\omega$ axis will yield oscillations, the amplitude of which is neither decaying nor growing with time. In practical cases, where noise is present, however, the amplitude of oscillations may increase at a rate determined by the noise power level. Therefore, a control system should not have closed-loop poles on the $j\omega$ axis.)

Note that the mere fact that all closed-loop poles lie in the left-half s plane does not guarantee satisfactory transient-response characteristics. If dominant complex-conjugate closed-loop poles lie close to the $j\omega$ axis, the transient response may exhibit excessive oscillations or may be very slow. Therefore, to guarantee fast, yet well-damped, transient-response characteristics, it is necessary that the closed-loop poles of the system lie in a particular region in the complex plane, such as the region bounded by the shaded area in Figure 5-23.

Since the relative stability and transient performance of a closed-loop control system are directly related to the closed-loop pole-zero configuration in the s plane, it is frequently necessary to adjust one or more system parameters in order to obtain suitable configurations. The effects of varying system parameters on the closed-loop poles will be discussed in detail in Chapter 6.

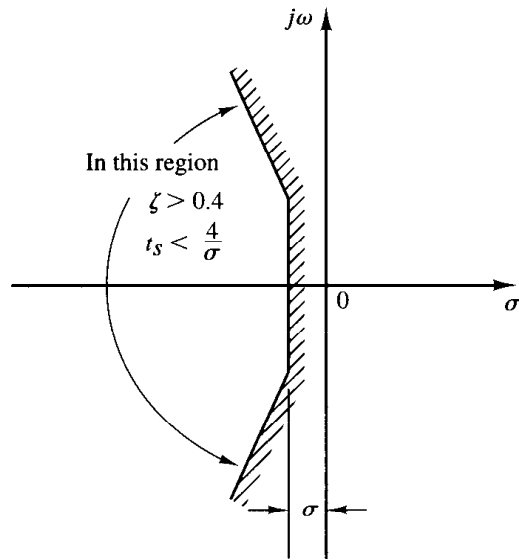


Figure 5-23

Region in the complex plane satisfying the conditions $\zeta > 0.4$ and $t_s < 4/\sigma$.

5-5 ROUTH'S STABILITY CRITERION

The most important problem in linear control systems concerns stability. That is, under what conditions will a system become unstable? If it is unstable, how should we stabilize the system? In Section 5-4 it was stated that a control system is stable if and only if all closed-loop poles lie in the left-half s plane. Since most linear closed-loop systems have closed-loop transfer functions of the form

$$\frac{C(s)}{R(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} = \frac{B(s)}{A(s)}$$

where the a 's and b 's are constants and $m \leq n$, we must first factor the polynomial $A(s)$ in order to find the closed-loop poles. A simple criterion, known as Routh's stability criterion, enables us to determine the number of closed-loop poles that lie in the right-half s plane without having to factor the polynomial.

Routh's stability criterion. Routh's stability criterion tells us whether or not there are unstable roots in a polynomial equation without actually solving for them. This stability criterion applies to polynomials with only a finite number of terms. When the criterion is applied to a control system, information about absolute stability can be obtained directly from the coefficients of the characteristic equation.

The procedure in Routh's stability criterion is as follows:

1. Write the polynomial in s in the following form:

$$a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0 \quad (5-6)$$

where the coefficients are real quantities. We assume that $a_n \neq 0$; that is, any zero root has been removed.

2. If any of the coefficients are zero or negative in the presence of at least one positive coefficient, there is a root or roots that are imaginary or that have positive real parts.

Therefore, in such a case, the system is not stable. If we are interested in only the absolute stability, there is no need to follow the procedure further. Note that all the coefficients must be positive. This is a necessary condition, as may be seen from the following argument: A polynomial in s having real coefficients can always be factored into linear and quadratic factors, such as $(s + a)$ and $(s^2 + bs + c)$, where a , b , and c are real. The linear factors yield real roots and the quadratic factors yield complex roots of the polynomial. The factor $(s^2 + bs + c)$ yields roots having negative real parts only if b and c are both positive. For all roots to have negative real parts, the constants a , b , c , and so on, in all factors must be positive. The product of any number of linear and quadratic factors containing only positive coefficients always yields a polynomial with positive coefficients. It is important to note that the condition that all the coefficients be positive is not sufficient to assure stability. The necessary but not sufficient condition for stability is that the coefficients of Equation (5-6) all be present and all have a positive sign. (If all a 's are negative, they can be made positive by multiplying both sides of the equation by -1 .)

3. If all coefficients are positive, arrange the coefficients of the polynomial in rows and columns according to the following pattern:

$$\begin{array}{cccccccc}
 s^n & a_0 & a_2 & a_4 & a_6 & . & . & . \\
 s^{n-1} & a_1 & a_3 & a_5 & a_7 & . & . & . \\
 s^{n-2} & b_1 & b_2 & b_3 & b_4 & . & . & . \\
 s^{n-3} & c_1 & c_2 & c_3 & c_4 & . & . & . \\
 s^{n-4} & d_1 & d_2 & d_3 & d_4 & . & . & . \\
 . & . & . & . & . & . & . & . \\
 . & . & . & . & . & . & . & . \\
 . & . & . & . & . & . & . & . \\
 s^2 & e_1 & e_2 & . & . & . & . & . \\
 s^1 & f_1 & . & . & . & . & . & . \\
 s^0 & g_1 & . & . & . & . & . & .
 \end{array}$$

The coefficients b_1 , b_2 , b_3 , and so on, are evaluated as follows:

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

.

.

.

The evaluation of the b 's is continued until the remaining ones are all zero. The same pattern of cross-multiplying the coefficients of the two previous rows is followed in evaluating the c 's, d 's, e 's, and so on. That is,

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

$$c_3 = \frac{b_1 a_7 - a_1 b_4}{b_1}$$

.

.

.

and

$$d_1 = \frac{c_1 b_2 - b_1 c_2}{c_1}$$

$$d_2 = \frac{c_1 b_3 - b_1 c_3}{c_1}$$

.

.

.

This process is continued until the n th row has been completed. The complete array of coefficients is triangular. Note that in developing the array an entire row may be divided or multiplied by a positive number in order to simplify the subsequent numerical calculation without altering the stability conclusion.

Routh's stability criterion states that the number of roots of Equation (5-6) with positive real parts is equal to the number of changes in sign of the coefficients of the first column of the array. It should be noted that the exact values of the terms in the first column need not be known; instead, only the signs are needed. The necessary and sufficient condition that all roots of Equation (5-6) lie in the left-half s plane is that all the coefficients of Equation (5-6) be positive and all terms in the first column of the array have positive signs.

EXAMPLE 5-1

Let us apply Routh's stability criterion to the following third-order polynomial:

$$a_0 s^3 + a_1 s^2 + a_2 s + a_3 = 0$$

where all the coefficients are positive numbers. The array of coefficients becomes

$$\begin{array}{ccc} s^3 & a_0 & a_2 \\ s^2 & a_1 & a_3 \\ s^1 & \frac{a_1 a_2 - a_0 a_3}{a_1} & \\ s^0 & a_3 & \end{array}$$

The condition that all roots have negative real parts is given by

$$a_1 a_2 > a_0 a_3$$

EXAMPLE 5-2

Consider the following polynomial:

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

Let us follow the procedure just presented and construct the array of coefficients. (The first two rows can be obtained directly from the given polynomial. The remaining terms are obtained from these. If any coefficients are missing, they may be replaced by zeros in the array.)

$$\begin{array}{cccc|cccc}
 s^4 & 1 & 3 & 5 & s^4 & 1 & 3 & 5 \\
 s^3 & 2 & 4 & 0 & s^3 & \cancel{2} & \cancel{4} & \emptyset \\
 & & & & & 1 & 2 & 0 \\
 s^2 & 1 & 5 & & s^2 & 1 & 5 & \\
 s^1 & -6 & & & s^1 & -3 & & \\
 s_0 & 5 & & & s^0 & 5 & &
 \end{array}$$

The second row is divided by 2.

In this example, the number of changes in sign of the coefficients in the first column is 2. This means that there are two roots with positive real parts. Note that the result is unchanged when the coefficients of any row are multiplied or divided by a positive number in order to simplify the computation.

Special cases. If a first-column term in any row is zero, but the remaining terms are not zero or there is no remaining term, then the zero term is replaced by a very small positive number ϵ and the rest of the array is evaluated. For example, consider the following equation:

$$s^3 + 2s^2 + s + 2 = 0 \quad (5-7)$$

The array of coefficients is

$$\begin{array}{ccc}
 s^3 & 1 & 1 \\
 s^2 & 2 & 2 \\
 s^1 & 0 \approx \epsilon & \\
 s^0 & 2 &
 \end{array}$$

If the sign of the coefficient above the zero (ϵ) is the same as that below it, it indicates that there are a pair of imaginary roots. Actually, Equation (5-7) has two roots at $s = \pm j$.

If, however, the sign of the coefficient above the zero (ϵ) is opposite that below it, it indicates that there is one sign change. For example, for the equation

$$s^3 - 3s + 2 = (s - 1)^2(s + 2) = 0$$

the array of coefficients is

$$\begin{array}{ccc}
 & s^3 & 1 & -3 \\
 \text{One sign change:} & \left\{ \begin{array}{l} s^2 \\ s^1 \end{array} \right. & \begin{array}{l} 0 \approx \epsilon \\ -3 - \frac{2}{\epsilon} \end{array} & \begin{array}{l} 2 \\ \frac{2}{\epsilon} \end{array} \\
 \text{One sign change:} & \left\{ \begin{array}{l} s^1 \\ s^0 \end{array} \right. & &
 \end{array}$$

There are two sign changes of the coefficients in the first column. This agrees with the correct result indicated by the factored form of the polynomial equation.

If all the coefficients in any derived row are zero, it indicates that there are roots of equal magnitude lying radially opposite in the s plane, that is, two real roots with equal magnitudes and opposite signs and/or two conjugate imaginary roots. In such a case, the evaluation of the rest of the array can be continued by forming an auxiliary polynomial

with the coefficients of the last row and by using the coefficients of the derivative of this polynomial in the next row. Such roots with equal magnitudes and lying radially opposite in the s plane can be found by solving the auxiliary polynomial, which is always even. For a $2n$ -degree auxiliary polynomial, there are n pairs of equal and opposite roots. For example, consider the following equation:

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

The array of coefficients is

$$\begin{array}{cccc} s^5 & 1 & 24 & -25 \\ s^4 & 2 & 48 & -50 \\ s^3 & 0 & 0 & \end{array} \quad \leftarrow \text{Auxiliary polynomial } P(s)$$

The terms in the s^3 row are all zero. The auxiliary polynomial is then formed from the coefficients of the s^4 row. The auxiliary polynomial $P(s)$ is

$$P(s) = 2s^4 + 48s^2 - 50$$

which indicates that there are two pairs of roots of equal magnitude and opposite sign. These pairs are obtained by solving the auxiliary polynomial equation $P(s) = 0$. The derivative of $P(s)$ with respect to s is

$$\frac{dP(s)}{ds} = 8s^3 + 96s$$

The terms in the s^3 row are replaced by the coefficients of the last equation, that is, 8 and 96. The array of coefficients then becomes

$$\begin{array}{cccc} s^5 & 1 & 24 & -25 \\ s^4 & 2 & 48 & -50 \\ s^3 & 8 & 96 & \\ s^2 & 24 & -50 & \\ s^1 & 112.7 & 0 & \\ s^0 & -50 & & \end{array} \quad \leftarrow \text{Coefficients of } dP(s)/ds$$

We see that there is one change in sign in the first column of the new array. Thus, the original equation has one root with a positive real part. By solving for roots of the auxiliary polynomial equation,

$$2s^4 + 48s^2 - 50 = 0$$

we obtain

$$s^2 = 1, \quad s^2 = -25$$

or

$$s = \pm 1, \quad s = \pm j5$$

These two pairs of roots are a part of the roots of the original equation. As a matter of fact, the original equation can be written in factored form as follows:

$$(s + 1)(s - 1)(s + j5)(s - j5)(s + 2) = 0$$

Clearly, the original equation has one root with a positive real part.

Relative stability analysis. Routh's stability criterion provides the answer to the question of absolute stability. This, in many practical cases, is not sufficient. We usually require information about the relative stability of the system. A useful approach for examining relative stability is to shift the s -plane axis and apply Routh's stability criterion. That is, we substitute

$$s = \hat{s} - \sigma \quad (\sigma = \text{constant})$$

into the characteristic equation of the system, write the polynomial in terms of $\hat{s}$; and apply Routh's stability criterion to the new polynomial in $\hat{s}$. The number of changes of sign in the first column of the array developed for the polynomial in $\hat{s}$ is equal to the number of roots that are located to the right of the vertical line $s = -\sigma$. Thus, this test reveals the number of roots that lie to the right of the vertical line $s = -\sigma$.

Application of Routh's stability criterion to control system analysis. Routh's stability criterion is of limited usefulness in linear control system analysis mainly because it does not suggest how to improve relative stability or how to stabilize an unstable system. It is possible, however, to determine the effects of changing one or two parameters of a system by examining the values that cause instability. In the following, we shall consider the problem of determining the stability range of a parameter value.

Consider the system shown in Figure 5–24. Let us determine the range of K for stability. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{K}{s(s^2 + s + 1)(s + 2) + K}$$

The characteristic equation is

$$s^4 + 3s^3 + 3s^2 + 2s + K = 0$$

The array of coefficients becomes

s^4	1	3	K
s^3	3	2	0
s^2	$\frac{7}{3}$	K	
s^1	$2 - \frac{9}{7}K$		
s^0	K		

For stability, K must be positive, and all coefficients in the first column must be positive. Therefore,

$$\frac{14}{9} > K > 0$$

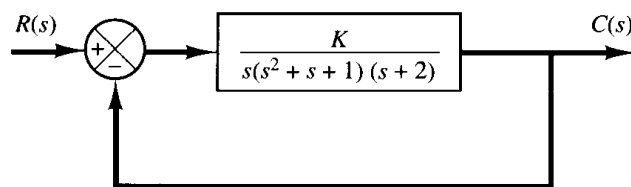


Figure 5–24
Control system.

When $K = \frac{14}{9}$, the system becomes oscillatory and, mathematically, the oscillation is sustained at constant amplitude.

5-6 PNEUMATIC CONTROLLERS

As the most versatile medium for transmitting signals and power, fluids, either as liquids or gases, have wide usage in industry. Liquids and gases can be distinguished basically by their relative incompressibilities and the fact that a liquid may have a free surface, whereas a gas expands to fill its vessel. In the engineering field the term *pneumatic* describes fluid systems that use air or gases and *hydraulic* applies to those using oil.

Pneumatic systems are extensively used in the automation of production machinery and in the field of automatic controllers. For instance, pneumatic circuits that convert the energy of compressed air into mechanical energy enjoy wide usage, and various types of pneumatic controllers are found in industry.

Since pneumatic systems and hydraulic systems are often compared, in what follows we shall give a brief comparison of these two kinds of systems.

Comparison between pneumatic systems and hydraulic systems. The fluid generally found in pneumatic systems is air; in hydraulic systems it is oil. And it is primarily the different properties of the fluids involved that characterize the differences between the two systems. These differences can be listed as follows:

1. Air and gases are compressible, whereas oil is incompressible.
2. Air lacks lubricating property and always contains water vapor. Oil functions as a hydraulic fluid as well as a lubricator.
3. The normal operating pressure of pneumatic systems is very much lower than that of hydraulic systems.
4. Output powers of pneumatic systems are considerably less than those of hydraulic systems.
5. Accuracy of pneumatic actuators is poor at low velocities, whereas accuracy of hydraulic actuators may be made satisfactory at all velocities.
6. In pneumatic systems, external leakage is permissible to a certain extent, but internal leakage must be avoided because the effective pressure difference is rather small. In hydraulic systems internal leakage is permissible to a certain extent, but external leakage must be avoided.
7. No return pipes are required in pneumatic systems when air is used, whereas they are always needed in hydraulic systems.
8. Normal operating temperature for pneumatic systems is 5° to 60°C (41° to 140°F). The pneumatic system, however, can be operated in the 0° to 200°C (32° to 392°F) range. Pneumatic systems are insensitive to temperature changes, in contrast to hydraulic systems, in which fluid friction due to viscosity depends greatly on temperature. Normal operating temperature for hydraulic systems is 20° to 70°C (68° to 158°F).
9. Pneumatic systems are fire- and explosion-proof, whereas hydraulic systems are not.

In what follows we begin with a mathematical modeling of pneumatic systems. Then we shall present pneumatic proportional controllers. We shall illustrate the fact that

proportional controllers utilize the principle of negative feedback in themselves. We shall give a detailed discussion of the principle by which proportional controllers operate. Finally, we shall treat methods for obtaining derivative and integral control actions. Throughout the discussions, we shall place emphasis on the fundamental principles, rather than on the details of the operation of the actual mechanisms.

Pneumatic systems. The past decades have seen a great development in low-pressure pneumatic controllers for industrial control systems, and today they are used extensively in industrial processes. Reasons for their broad appeal include an explosion-proof character, simplicity, and ease of maintenance.

Resistance and capacitance of pressure systems. Many industrial processes and pneumatic controllers involve the flow of a gas or air through connected pipelines and pressure vessels.

Consider the pressure system shown in Figure 5-25(a). The gas flow through the restriction is a function of the gas pressure difference $p_i - p_o$. Such a pressure system may be characterized in terms of a resistance and a capacitance.

The gas flow resistance R may be defined as follows:

$$R = \frac{\text{change in gas pressure difference, lb}_f/\text{ft}^2}{\text{change in gas flow rate, lb/sec}}$$

or

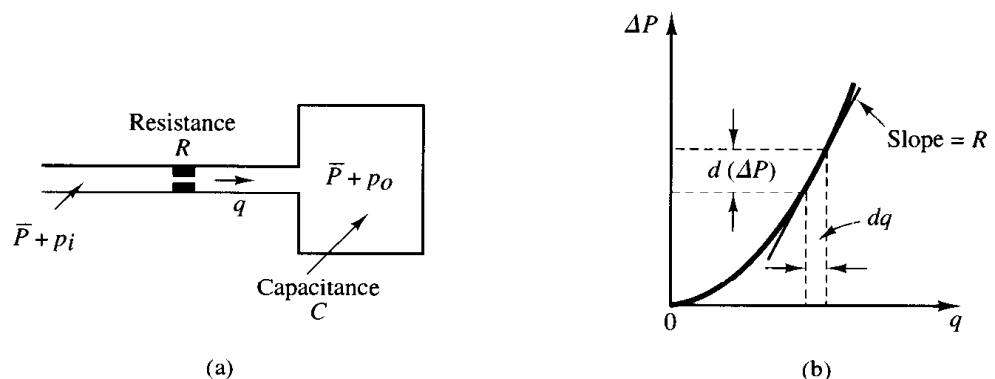
$$R = \frac{d(\Delta P)}{dq} \quad (5-8)$$

where $d(\Delta P)$ is a small change in the gas pressure difference and dq is a small change in the gas flow rate. Computation of the value of the gas flow resistance R may be quite time consuming. Experimentally, however, it can be easily determined from a plot of the pressure difference versus flow rate by calculating the slope of the curve at a given operating condition, as shown in Figure 5-25(b).

The capacitance of the pressure vessel may be defined by

$$C = \frac{\text{change in gas stored, lb}}{\text{change in gas pressure, lb}_f/\text{ft}^2}$$

Figure 5-25
(a) Schematic diagram of a pressure system; (b) pressure difference versus flow rate curve.



or

$$C = \frac{dm}{dp} = V \frac{d\rho}{dp} \quad (5-9)$$

where C = capacitance, lb-ft²/lb_f
 m = mass of gas in vessel, lb
 p = gas pressure, lb_f/ft²
 V = volume of vessel, ft³
 ρ = density, lb/ft³

The capacitance of the pressure system depends on the type of expansion process involved. The capacitance can be calculated by use of the ideal gas law. If the gas expansion process is polytropic and the change of state of the gas is between isothermal and adiabatic, then

$$p \left(\frac{V}{m} \right)^n = \frac{p}{\rho^n} = \text{constant} \quad (5-10)$$

where n = polytropic exponent.

For ideal gases,

$$p\bar{v} = \bar{R}T \quad \text{or} \quad pv = \frac{\bar{R}}{M} T$$

where p = absolute pressure, lb_f/ft²
 $\bar{v}$ = volume occupied by 1 mole of a gas, ft³/lb-mole
 $\bar{R}$ = universal gas constant, ft-lb_f/lb-mole °R
 T = absolute temperature, °R
 v = specific volume of gas, ft³/lb
 M = molecular weight of gas per mole, lb/lb-mole

Thus

$$pv = \frac{p}{\rho} = \frac{\bar{R}}{M} T = R_{\text{gas}} T \quad (5-11)$$

where R_{gas} = gas constant, ft-lb_f/lb °R.

The polytropic exponent n is unity for isothermal expansion. For adiabatic expansion, n is equal to the ratio of specific heats c_p/c_v , where c_p is the specific heat at constant pressure and c_v is the specific heat at constant volume. In many practical cases, the value of n is approximately constant, and thus the capacitance may be considered constant. The value of $d\rho/dp$ is obtained from Equations (5-10) and (5-11) as

$$\frac{d\rho}{dp} = \frac{1}{nR_{\text{gas}}T}$$

The capacitance is then obtained as

$$C = \frac{V}{nR_{\text{gas}}T} \quad (5-12)$$

The capacitance of a given vessel is constant if the temperature stays constant. (In many practical cases, the polytropic exponent n is approximately 1.0 ~ 1.2 for gases in uninsulated metal vessels.)

Pressure systems. Consider the system shown in Figure 5-25(a). If we assume only small deviations in the variables from their respective steady-state values, then this system may be considered linear.

Let us define

$\bar{P}$ = gas pressure in the vessel at steady state (before changes in pressure have occurred), lb_f/ft²

p_i = small change in inflow gas pressure, lb_f/ft²

p_o = small change in gas pressure in the vessel, lb_f/ft²

V = volume of the vessel, ft³

m = mass of gas in vessel, lb

q = gas flow rate, lb/sec

ρ = density of gas, lb/ft³

For small values of p_i and p_o , the resistance R given by Equation (5-8) becomes constant and may be written as

$$R = \frac{p_i - p_o}{q}$$

The capacitance C is given by Equation (5-9), or

$$C = \frac{dm}{dp}$$

Since the pressure change dp_o times the capacitance C is equal to the gas added to the vessel during dt seconds, we obtain

$$C dp_o = q dt$$

or

$$C \frac{dp_o}{dt} = \frac{p_i - p_o}{R}$$

which can be written as

$$RC \frac{dp_o}{dt} + p_o = p_i$$

If p_i and p_o are considered the input and output, respectively, then the transfer function of the system is

$$\frac{P_o(s)}{P_i(s)} = \frac{1}{RCs + 1}$$

where RC has the dimension of time and is the time constant of the system.

Pneumatic nozzle-flapper amplifiers. A schematic diagram of a pneumatic nozzle-flapper amplifier is shown in Figure 5-26(a). The power source for this amplifier is a supply of air at constant pressure. The nozzle-flapper amplifier converts small changes in the position of the flapper into large changes in the back pressure in the nozzle. Thus a large power output can be controlled by the very little power that is needed to position the flapper.

In Figure 5-26(a), pressurized air is fed through the orifice, and the air is ejected from the nozzle toward the flapper. Generally, the supply pressure P_s for such a controller is 20 psig (1.4 kg/cm² gage). The diameter of the orifice is on the order of 0.01 in. (0.25 mm) and that of the nozzle is on the order of 0.016 in. (0.4 mm). To ensure proper functioning of the amplifier, the nozzle diameter must be larger than the orifice diameter.

In operating this system, the flapper is positioned against the nozzle opening. The nozzle back pressure P_b is controlled by the nozzle-flapper distance X . As the flapper approaches the nozzle, the opposition to the flow of air through the nozzle increases, with the result that the nozzle back pressure P_b increases. If the nozzle is completely closed by the flapper, the nozzle back pressure P_b becomes equal to the supply pressure P_s . If the flapper is moved away from the nozzle, so that the nozzle-flapper distance is wide (on the order of 0.01 in.), then there is practically no restriction to flow, and the nozzle back pressure P_b takes on a minimum value that depends on the nozzle-flapper device. (The lowest possible pressure will be the ambient pressure P_a .)

Note that, because the air jet puts a force against the flapper, it is necessary to make the nozzle diameter as small as possible.

A typical curve relating the nozzle back pressure P_b to the nozzle-flapper distance X is shown in Figure 5-26(b). The steep and almost linear part of the curve is utilized in the actual operation of the nozzle-flapper amplifier. Because the range of flapper displacements is restricted to a small value, the change in output pressure is also small, unless the curve is very steep.

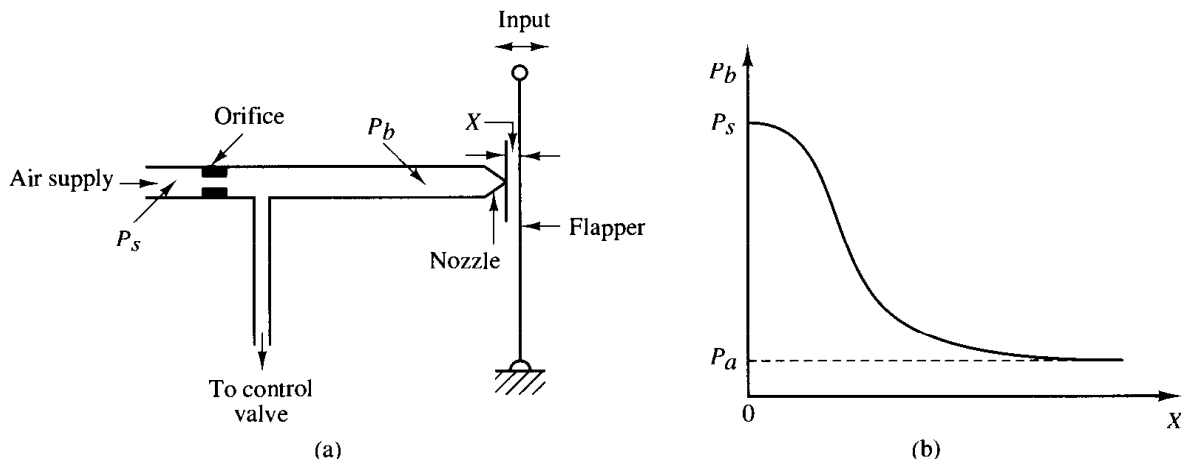


Figure 5-26

(a) Schematic diagram of a pneumatic nozzle-flapper amplifier; (b) characteristic curve relating nozzle back pressure and nozzle-flapper distance.

The nozzle-flapper amplifier converts displacement into a pressure signal. Since industrial process control systems require large output power to operate large pneumatic actuating valves, the power amplification of the nozzle-flapper amplifier is usually insufficient. Consequently, a pneumatic relay often serves as a power amplifier in connection with the nozzle-flapper amplifier.

Pneumatic relays. In practice, in a pneumatic controller, a nozzle-flapper amplifier acts as the first-stage amplifier and a pneumatic relay as the second-stage amplifier. The pneumatic relay is capable of handling a large quantity of airflow.

A schematic diagram of a pneumatic relay is shown in Figure 5-27(a). As the nozzle back pressure P_b increases, the diaphragm valve moves downward. The opening to the atmosphere decreases and the opening to the pneumatic valve increases, thereby increasing the control pressure P_c . When the diaphragm valve closes the opening to the atmosphere, the control pressure P_c becomes equal to the supply pressure P_s . When the nozzle back pressure P_b decreases and the diaphragm valve moves upward and shuts off the air supply, the control pressure P_c drops to the ambient pressure P_a . The control pressure P_c can thus be made to vary from 0 psig to full supply pressure, usually 20 psig.

The total movement of the diaphragm valve is very small. In all positions of the valve, except at the position to shut off the air supply, air continues to bleed into the atmosphere, even after the equilibrium condition is attained between the nozzle back pressure and the control pressure. Thus the relay shown in Figure 5-27(a) is called a bleed-type relay.

There is another type of relay, the nonbleed type. In this one the air bleed stops when the equilibrium condition is obtained and, therefore, there is no loss of pressurized air at steady-state operation. Note, however, that the nonbleed-type relay must have an atmospheric relief to release the control pressure P_c from the pneumatic actuating valve. A schematic diagram of a nonbleed-type relay is shown in Figure 5-27(b).

In either type of relay, the air supply is controlled by a valve, which is in turn controlled by the nozzle back pressure. Thus, the nozzle back pressure is converted into the control pressure with power amplification.

Since the control pressure P_c changes almost instantaneously with changes in the nozzle back pressure P_b , the time constant of the pneumatic relay is negligible compared with the other larger time constants of the pneumatic controller and the plant.

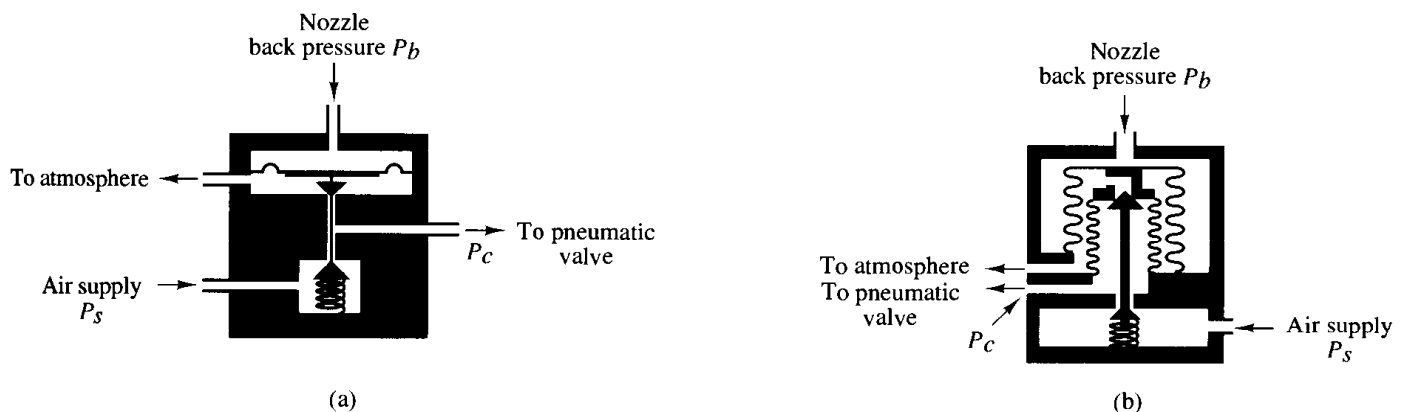


Figure 5-27

(a) Schematic diagram of a bleed-type relay; (b) schematic diagram of a nonbleed-type relay.